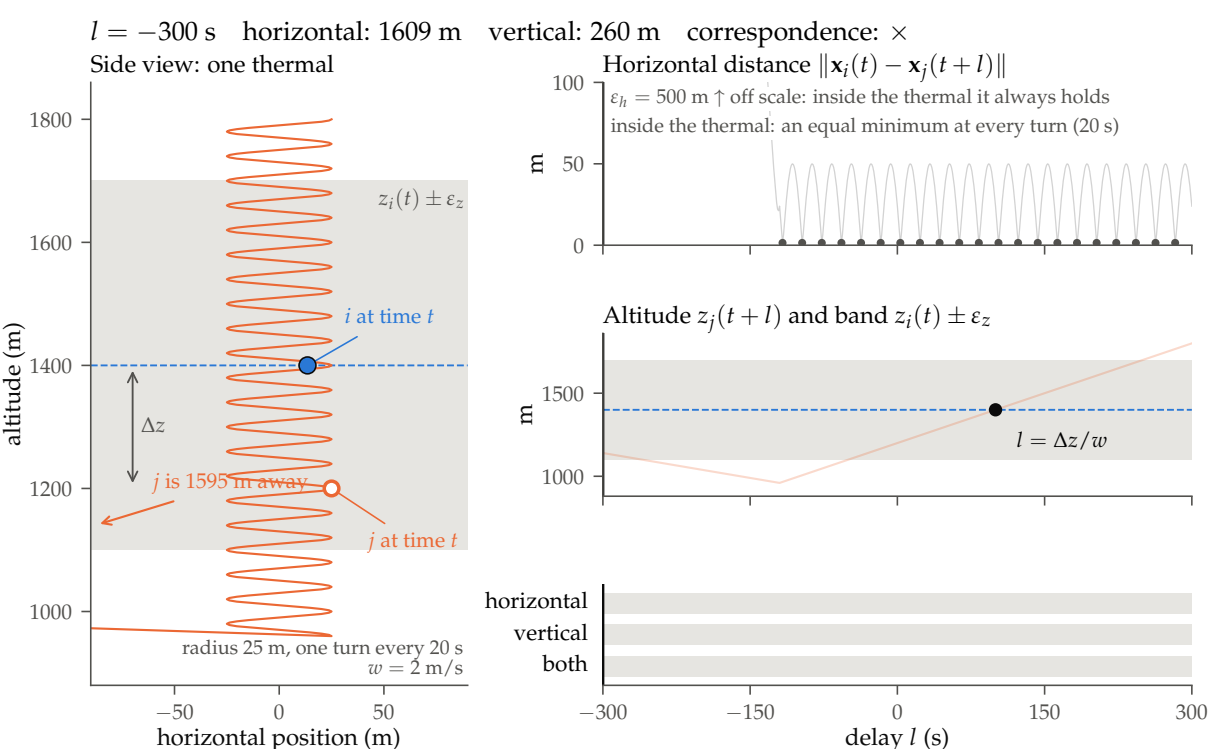
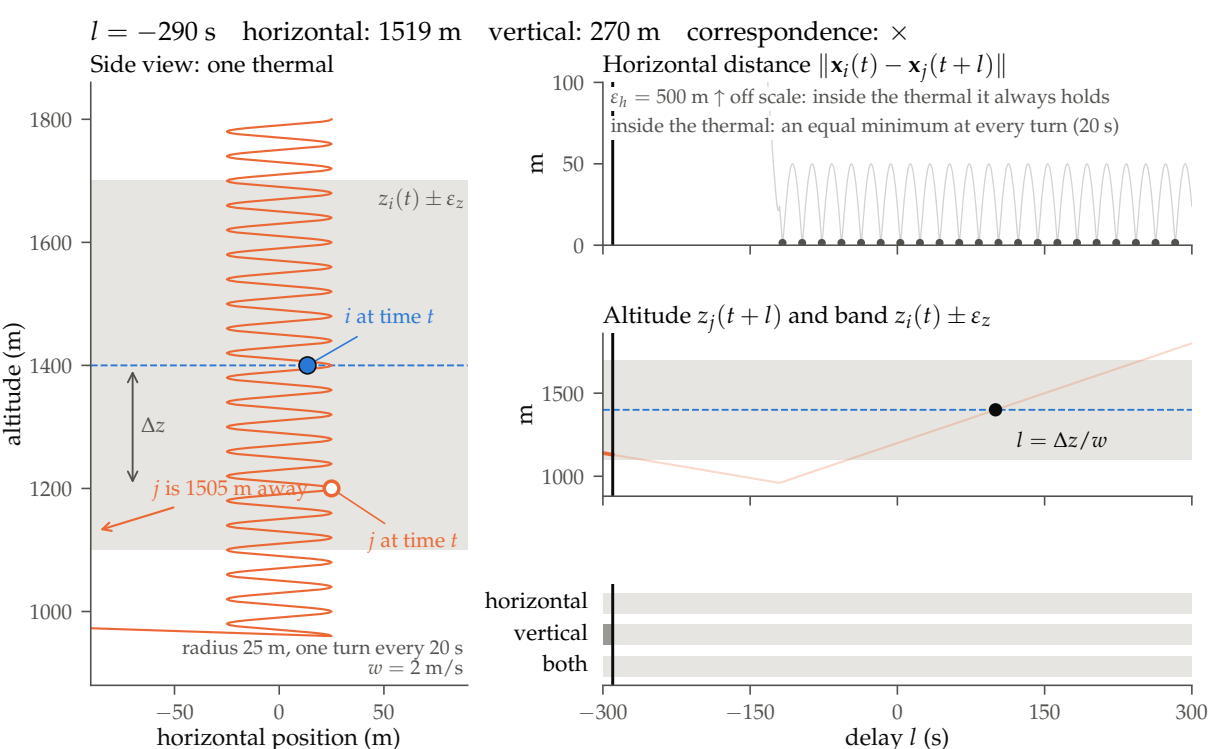
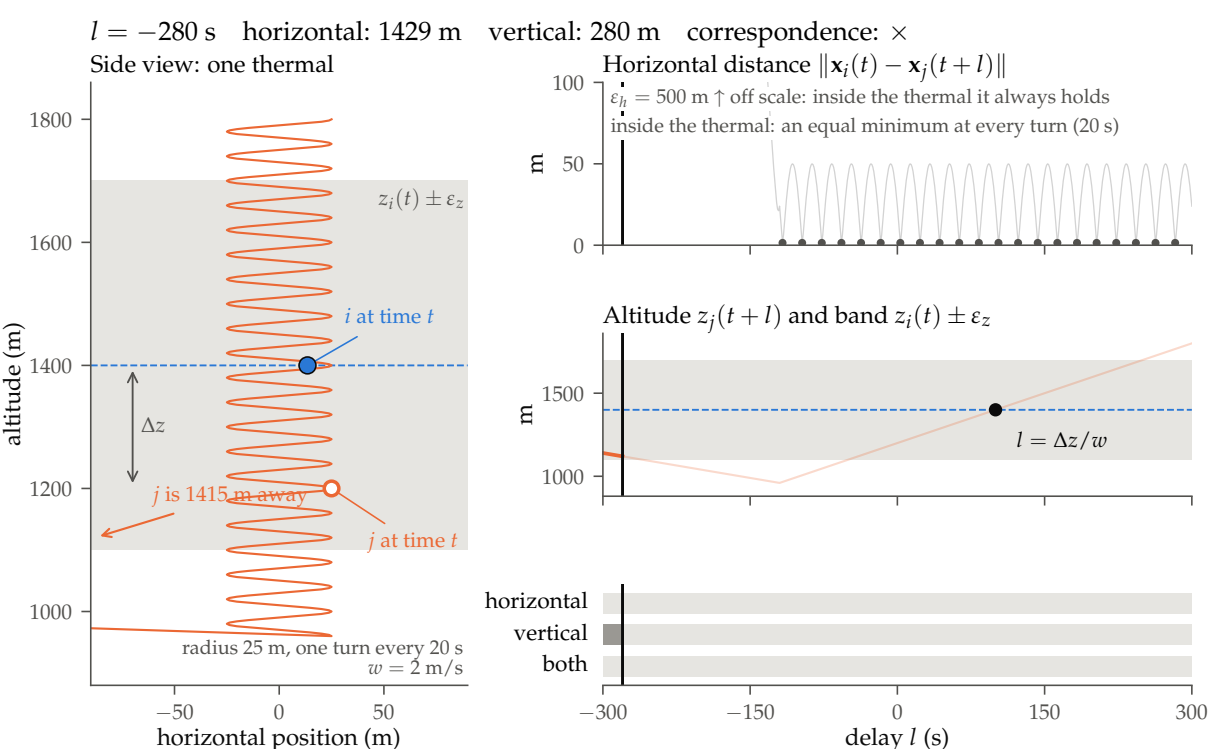
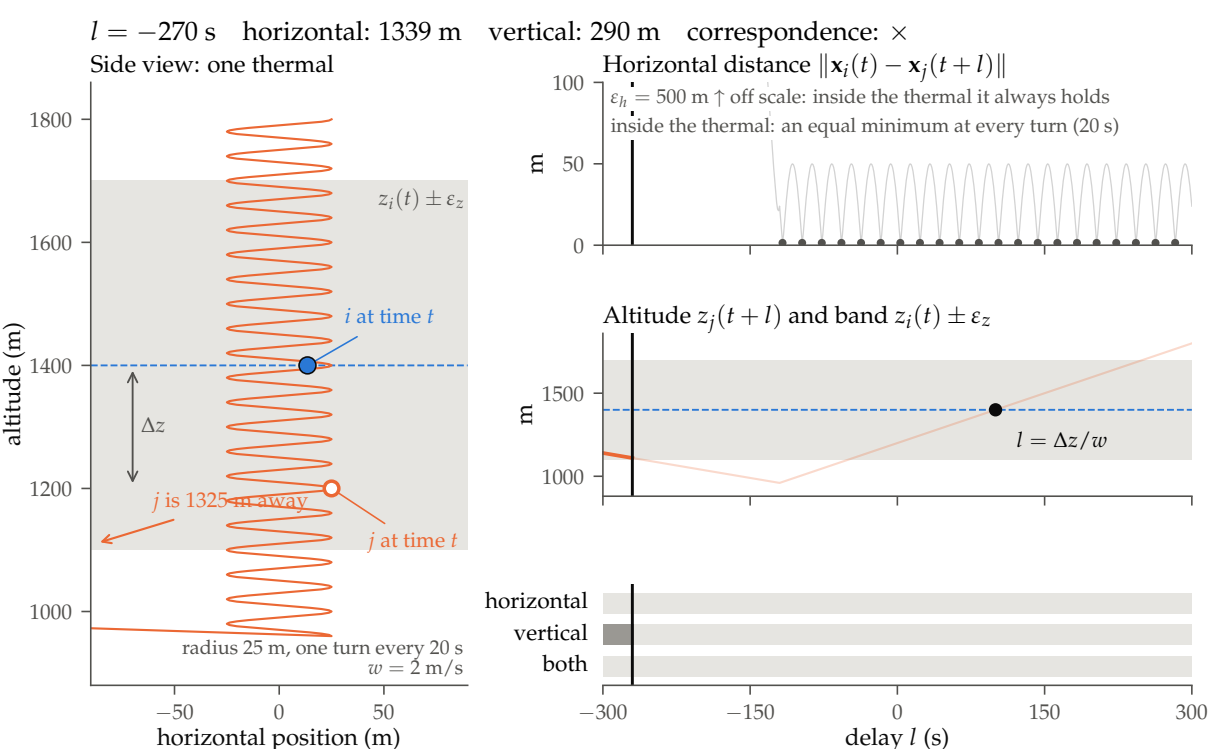
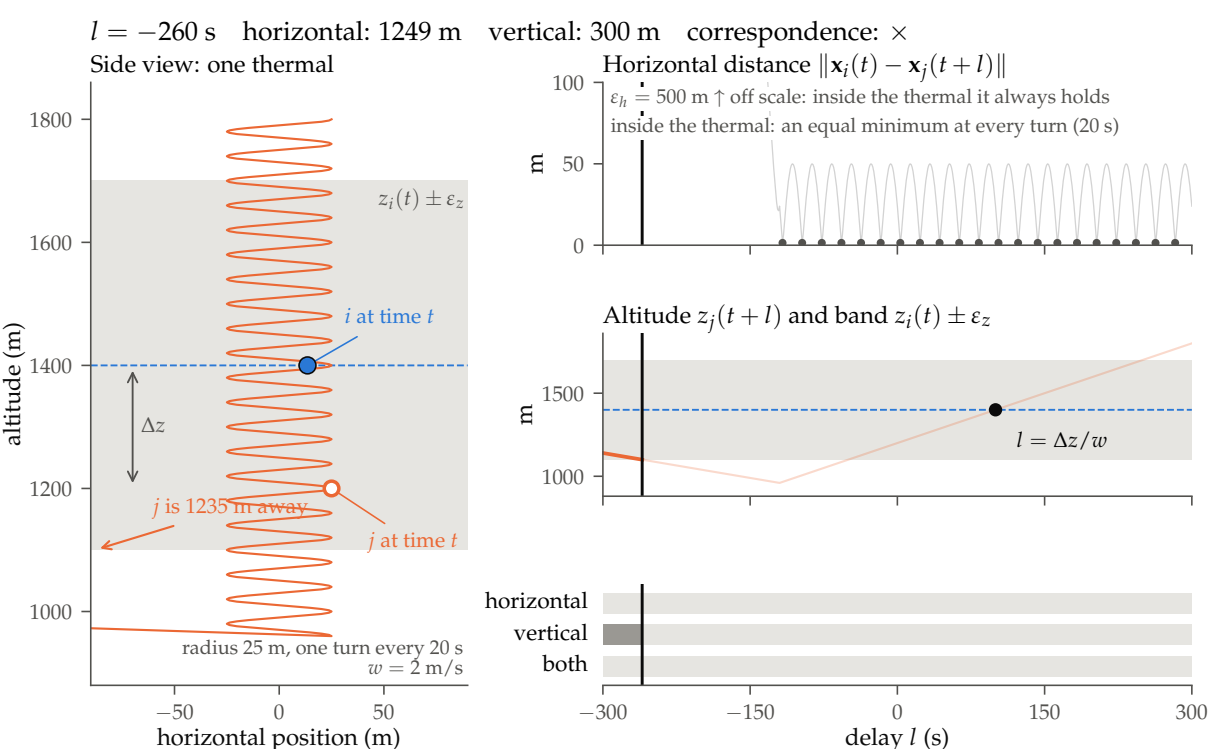
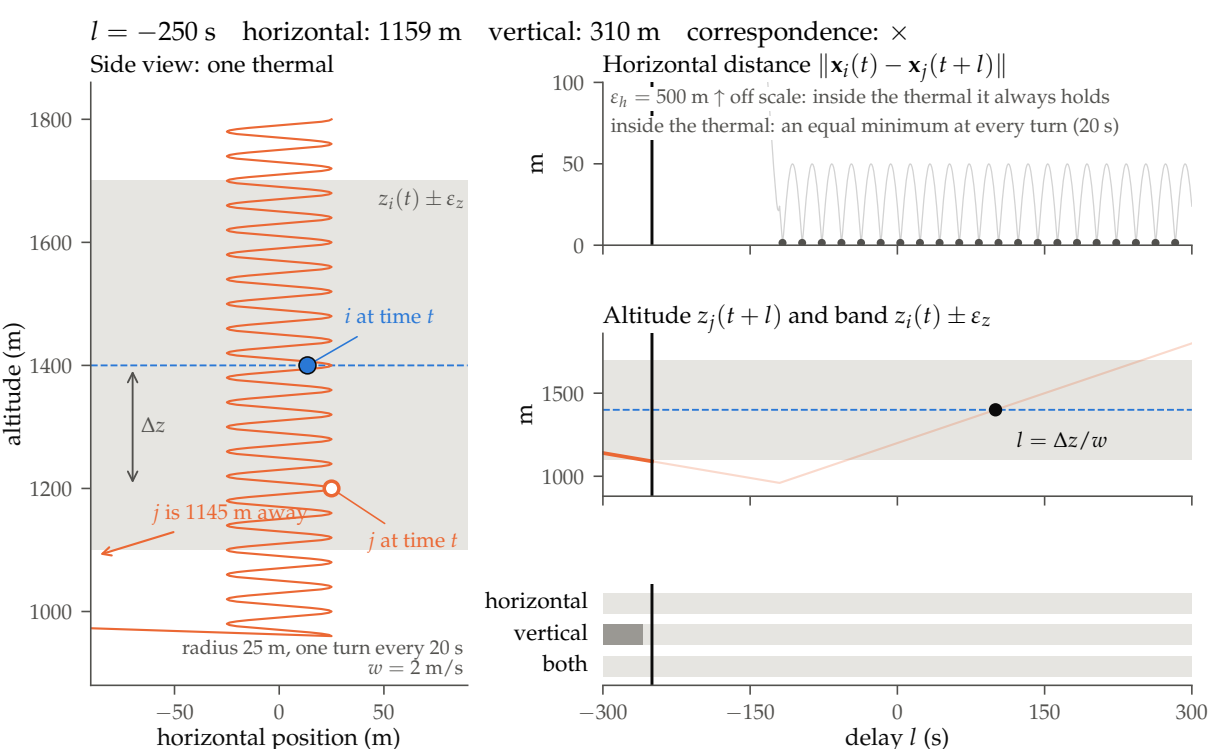

$l = -300$ s horizontal: 1609 m vertical: 260 m correspondence: ×
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j is 1595 m away
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

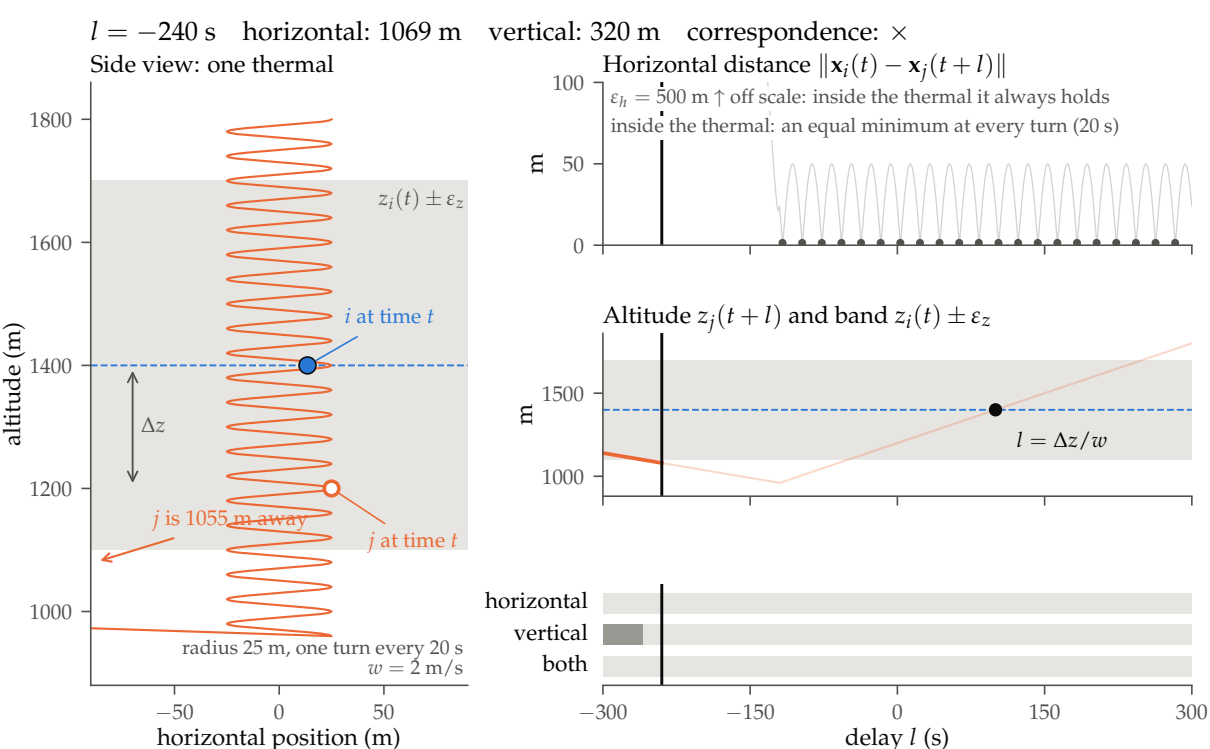

$l = -290$ s horizontal: 1519 m vertical: 270 m correspondence: ×
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j is 1505 m away
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

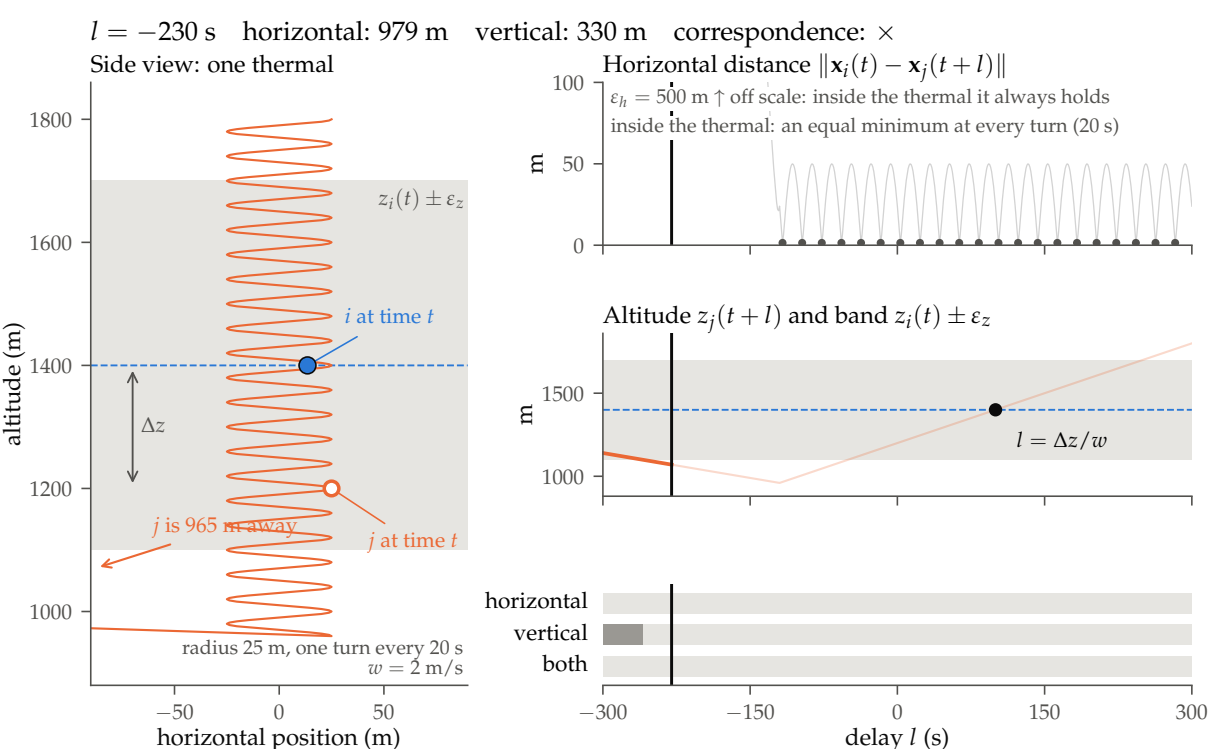

$l = -280$ s horizontal: 1429 m vertical: 280 m correspondence: ×
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j is 1415 m away
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

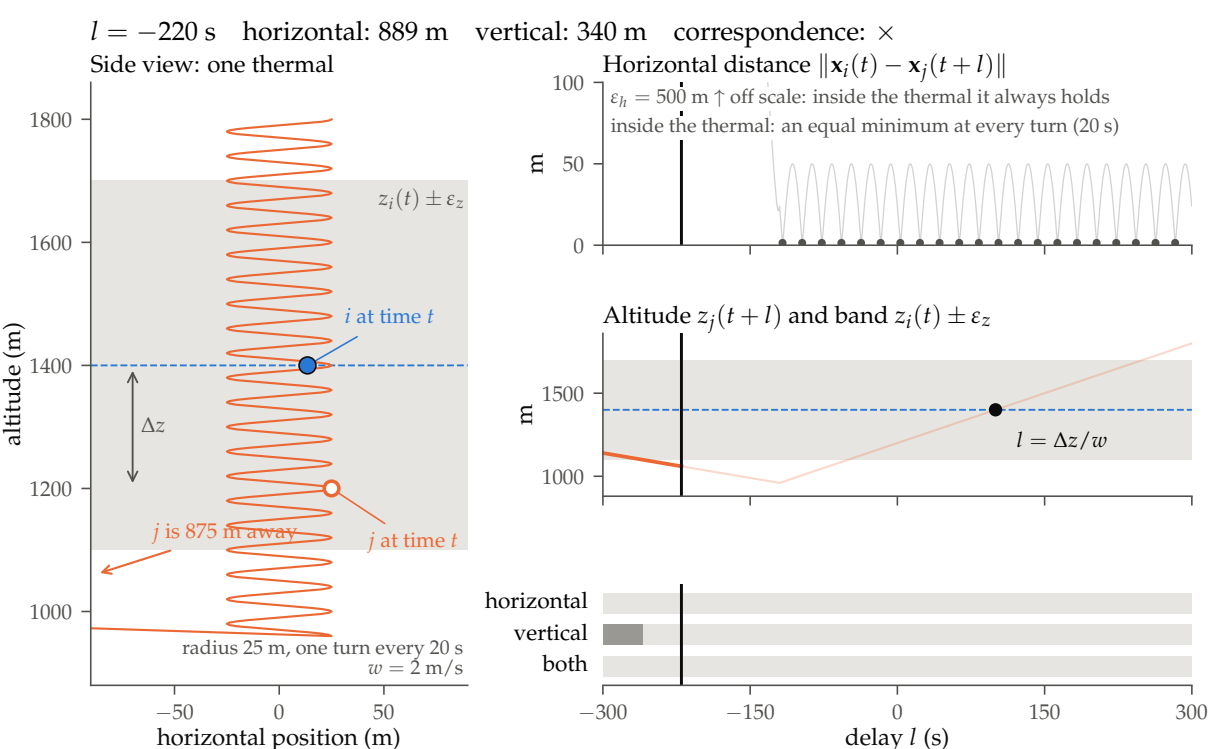

$l = -270$ s horizontal: 1339 m vertical: 290 m correspondence: ×
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j is 1325 m away
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

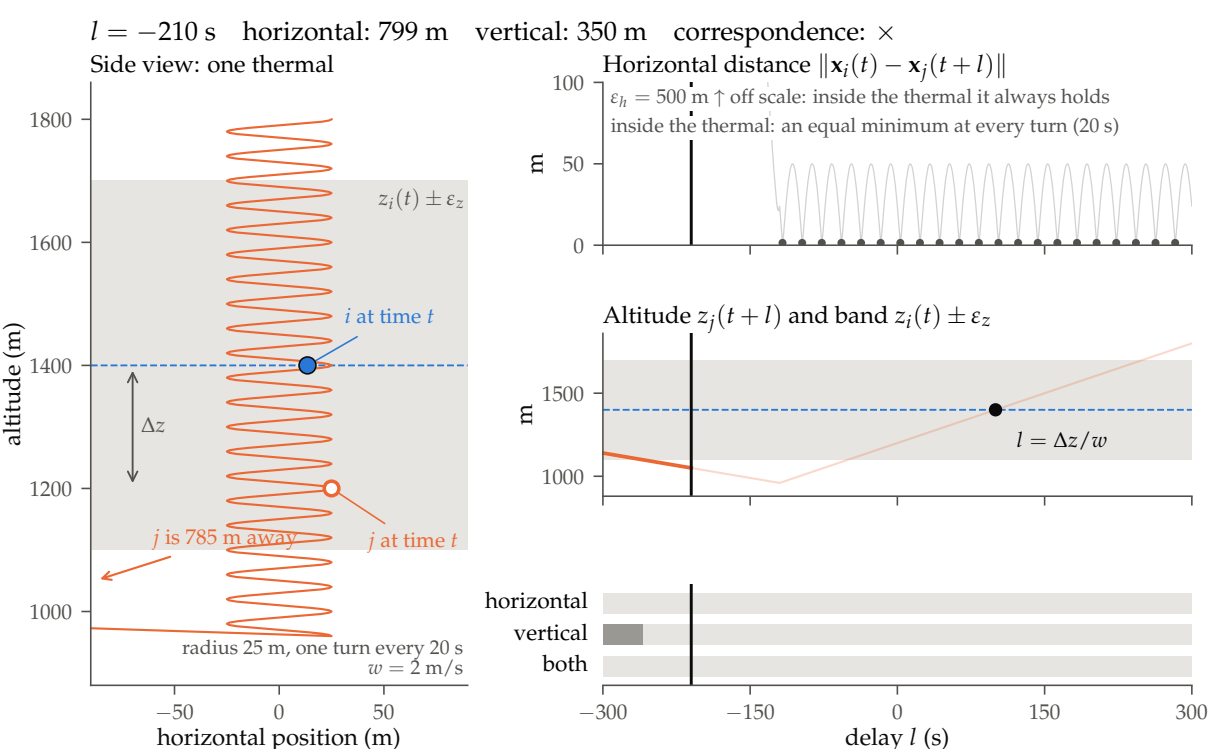

$l = -260$ s horizontal: 1249 m vertical: 300 m correspondence: ×
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j is 1235 m away
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

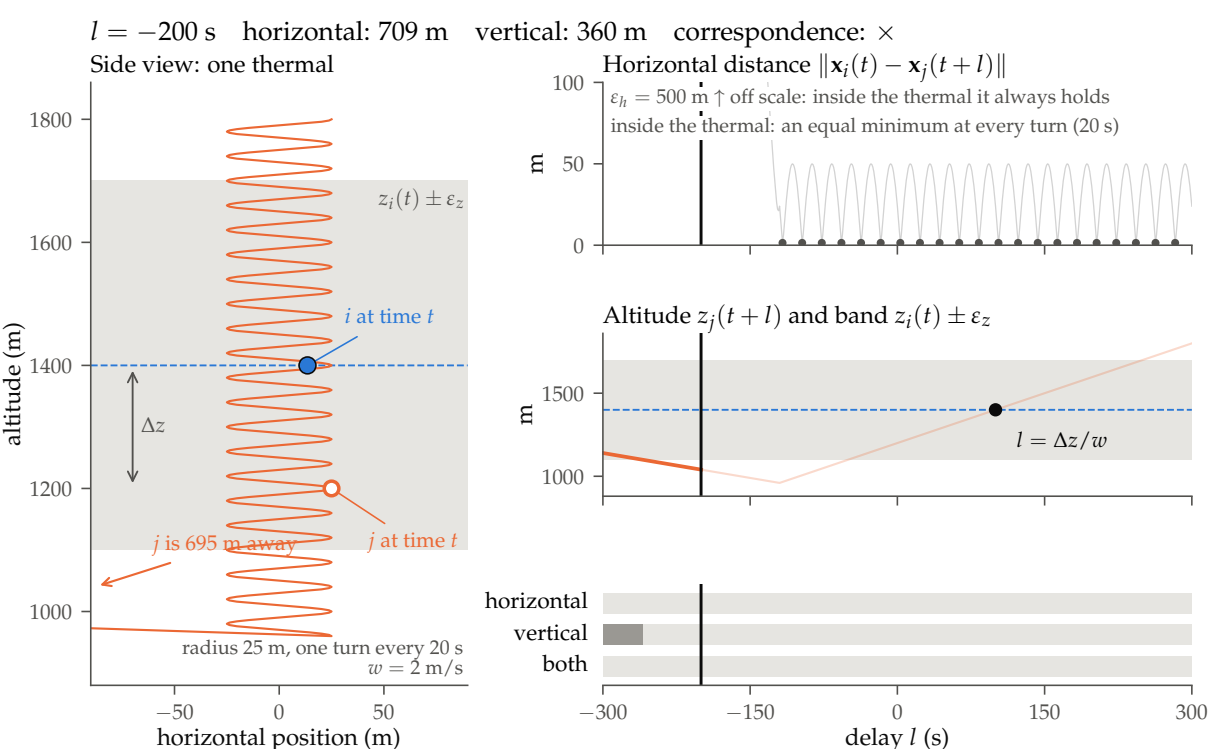

$l = -250$ s horizontal: 1159 m vertical: 310 m correspondence: ×
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j is 1145 m away
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

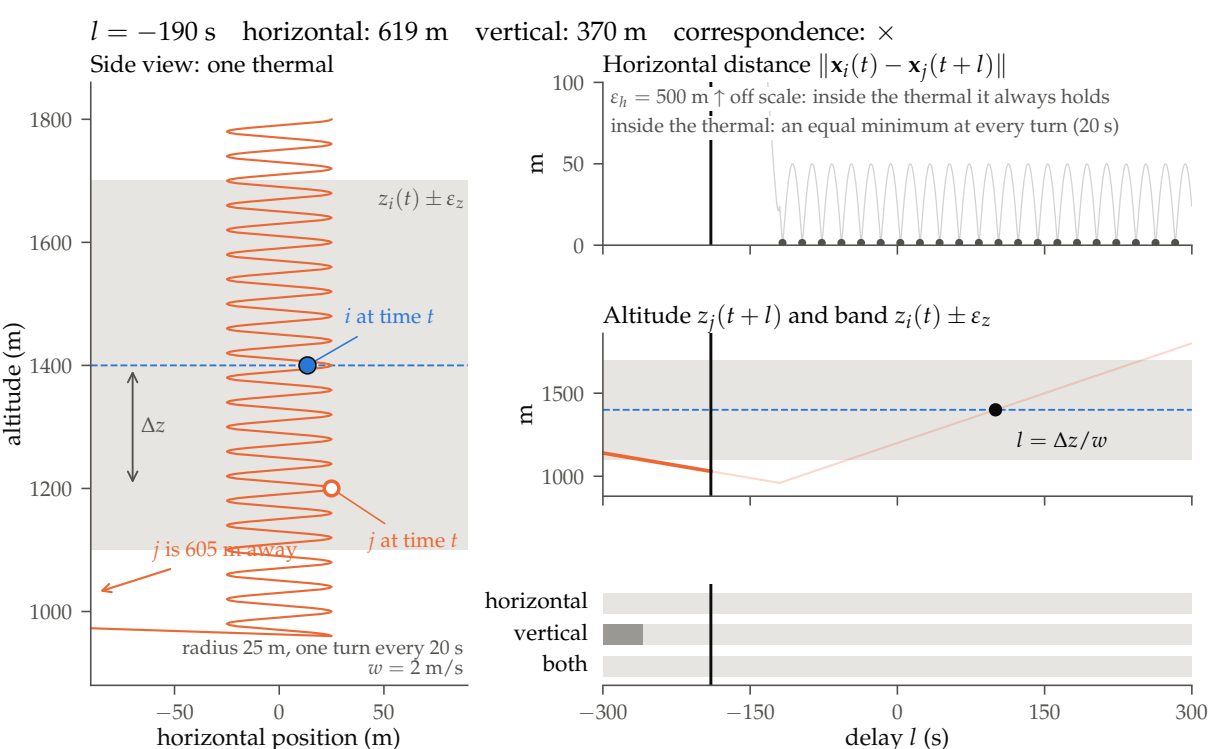

$l = -240$ s horizontal: 1069 m vertical: 320 m correspondence: ×
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j is 1055 m away
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

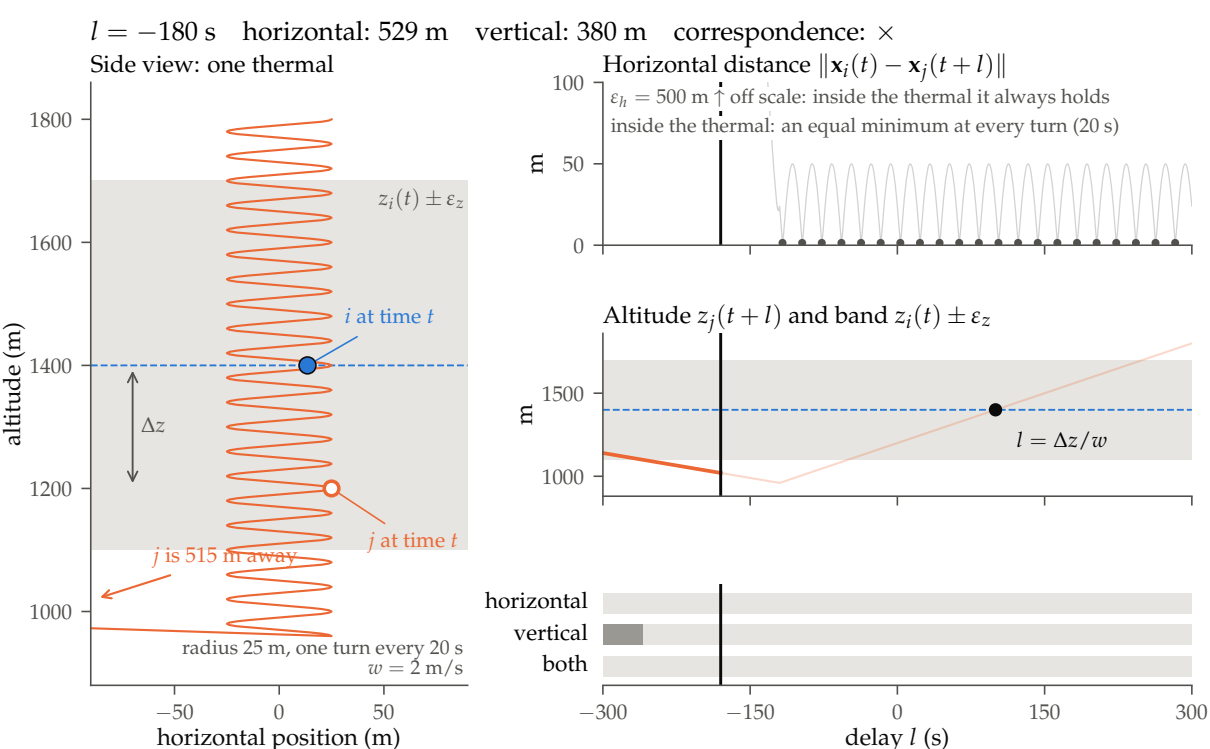

$l = -230$ s horizontal: 979 m vertical: 330 m correspondence: ×
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j is 965 m away
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

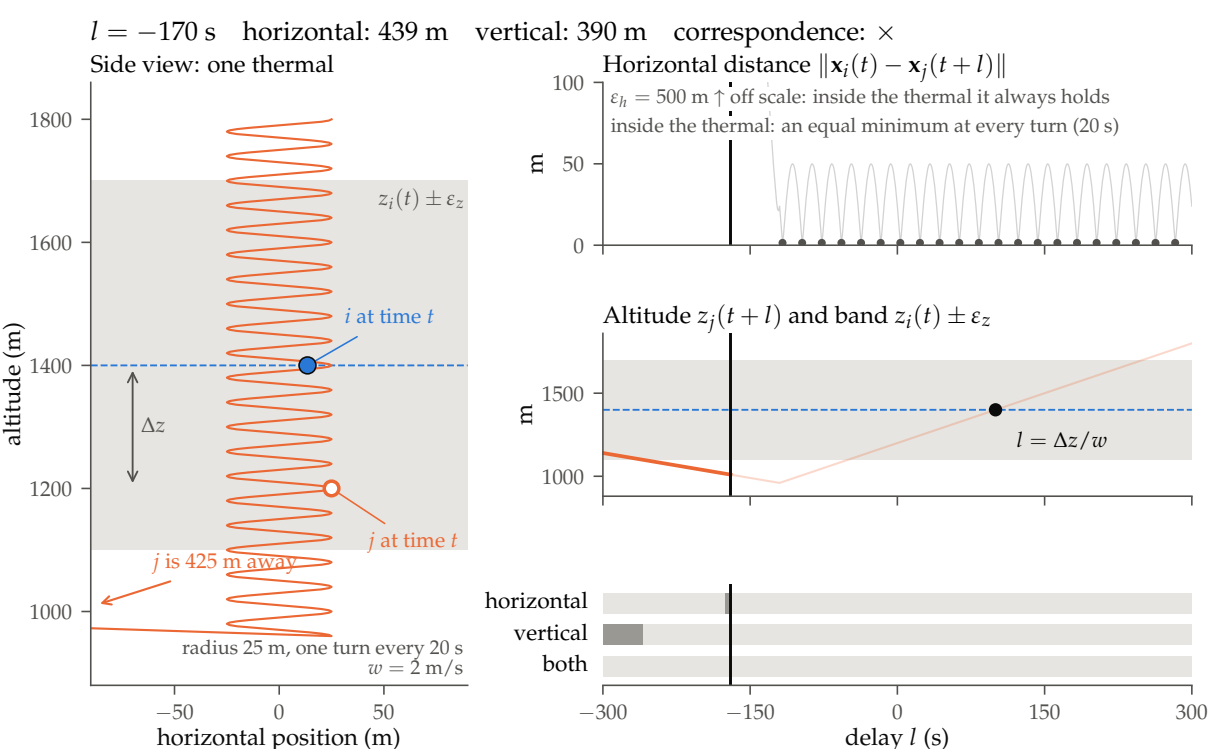

$l = -220$ s horizontal: 889 m vertical: 340 m correspondence: ×
Side view: one thermal
altitude (m)
1800
1600
1400
1200
1000
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j is 875 m away
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
m
100
50
0
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
m
1500
1000
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

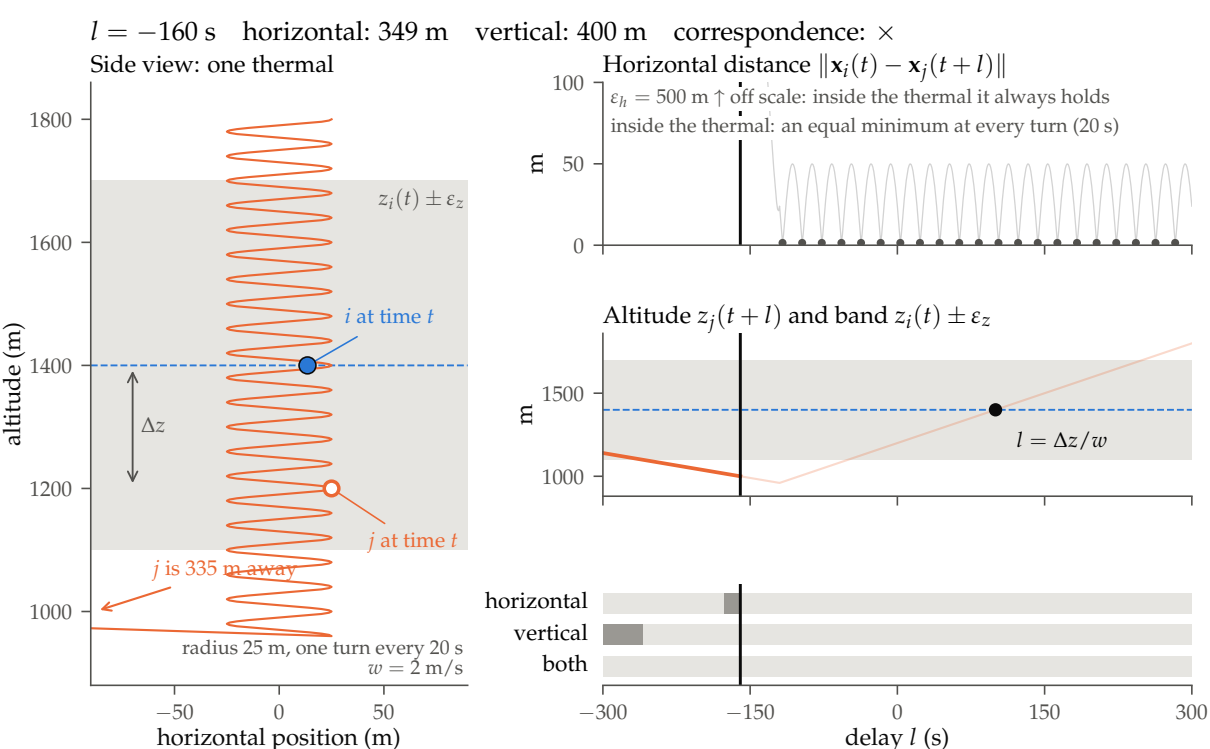

$l = -210$ s horizontal: 799 m vertical: 350 m correspondence: ×
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j is 785 m away
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
100
50
0
m
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

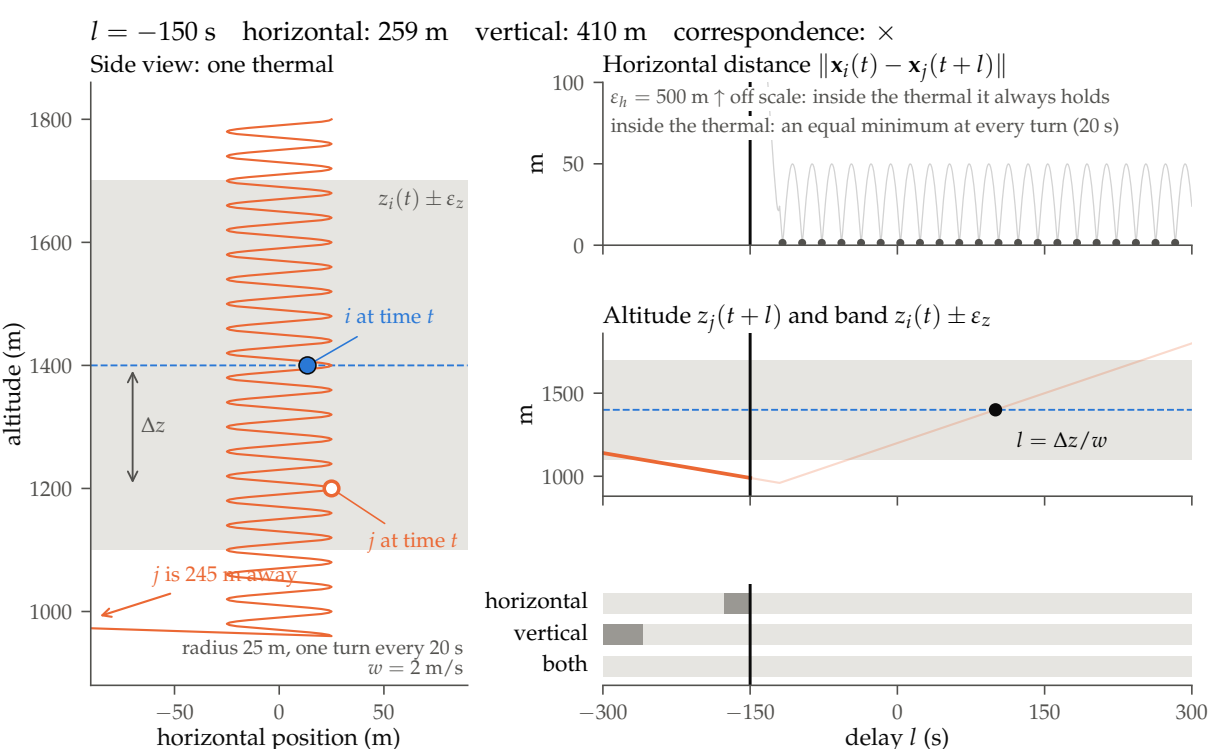

$l = -200$ s horizontal: 709 m vertical: 360 m correspondence: ×
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j is 695 m away
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

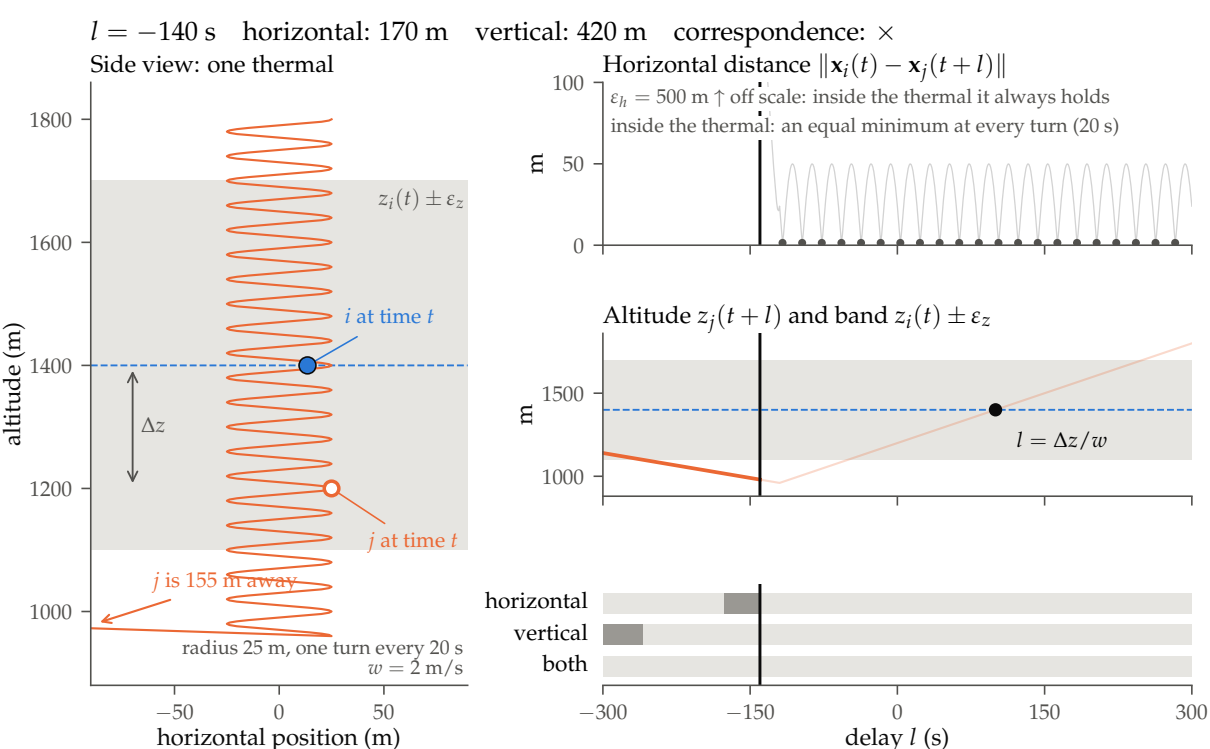

$l = -190$ s horizontal: 619 m vertical: 370 m correspondence: ×
Side view: one thermal
altitude (m)
1800
1600
1400
1200
1000
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
j is 605 m away
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

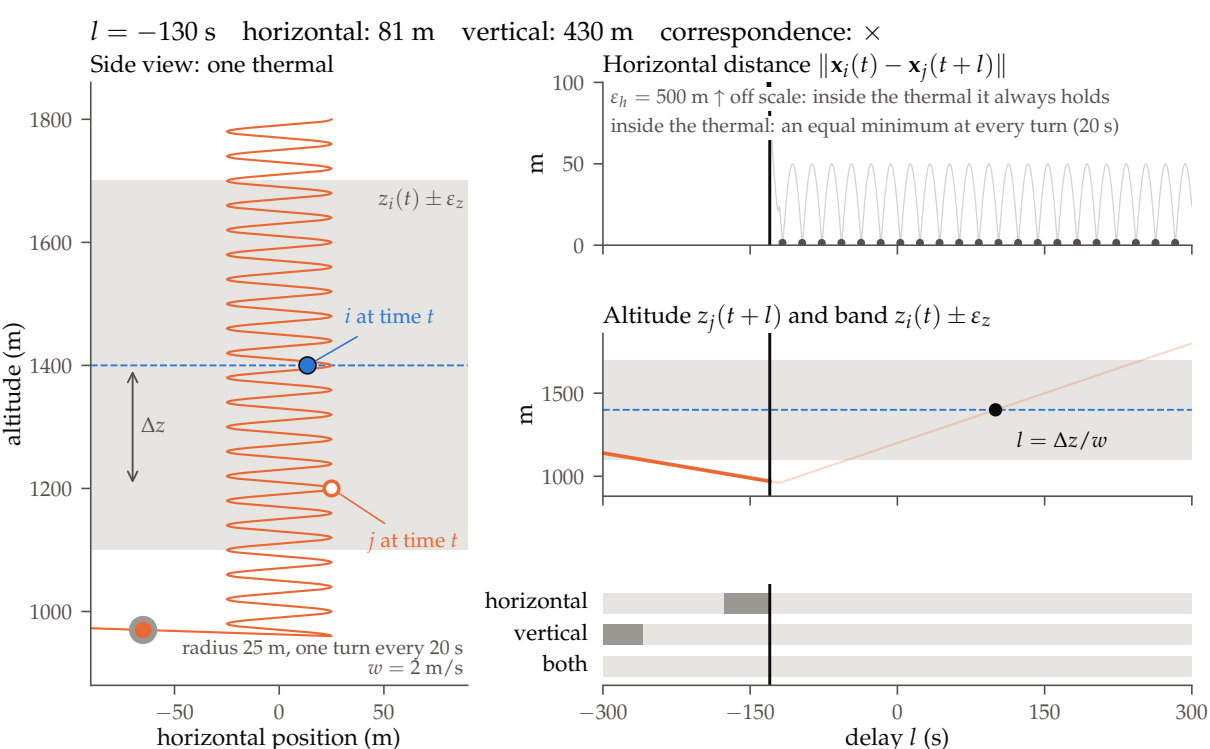

$l = -180$ s horizontal: 529 m vertical: 380 m correspondence: ×
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
j is 515 m away
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

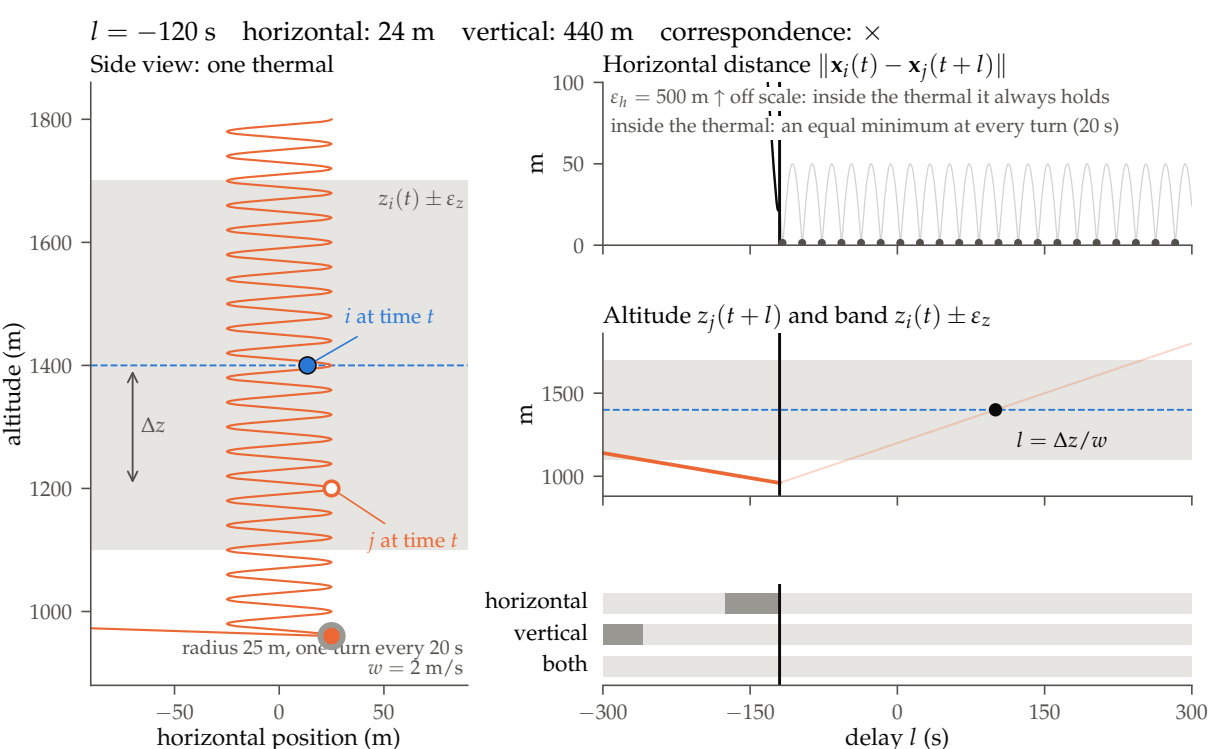

$l = -170$ s horizontal: 439 m vertical: 390 m correspondence: ×
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
j is 425 m away
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

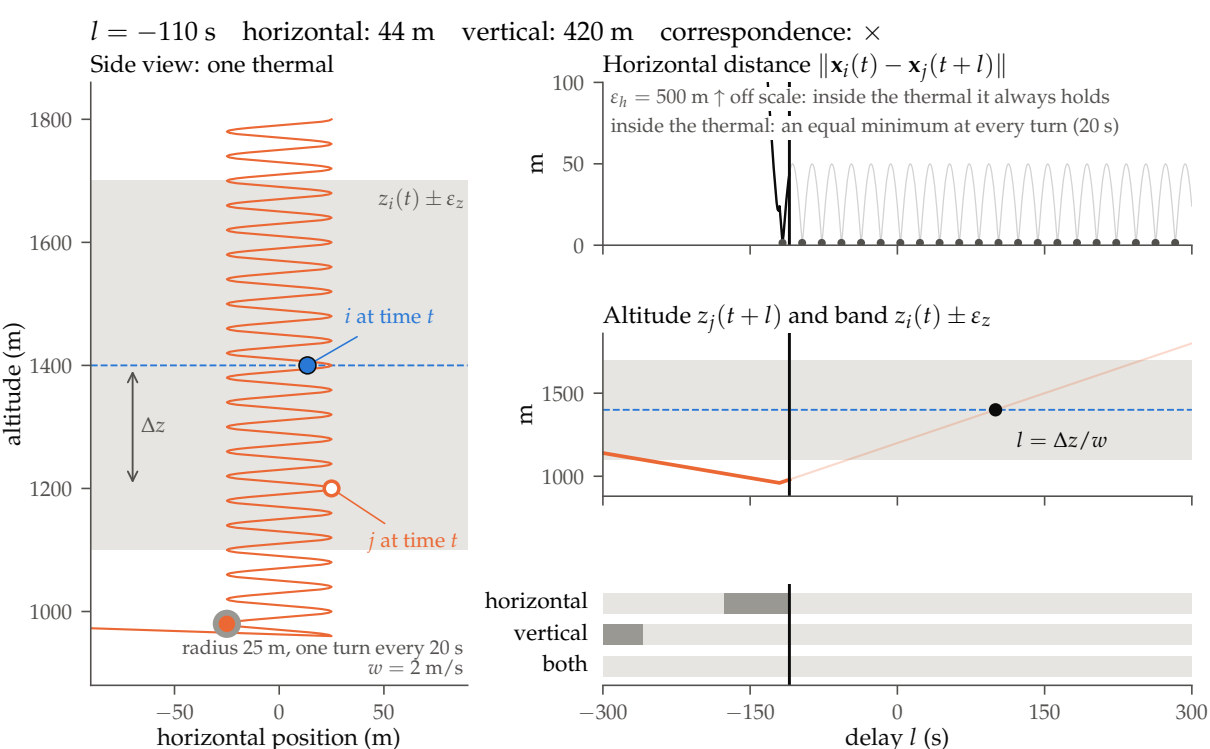

$l = -160$ s horizontal: 349 m vertical: 400 m correspondence: ×
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
j is 335 m away
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

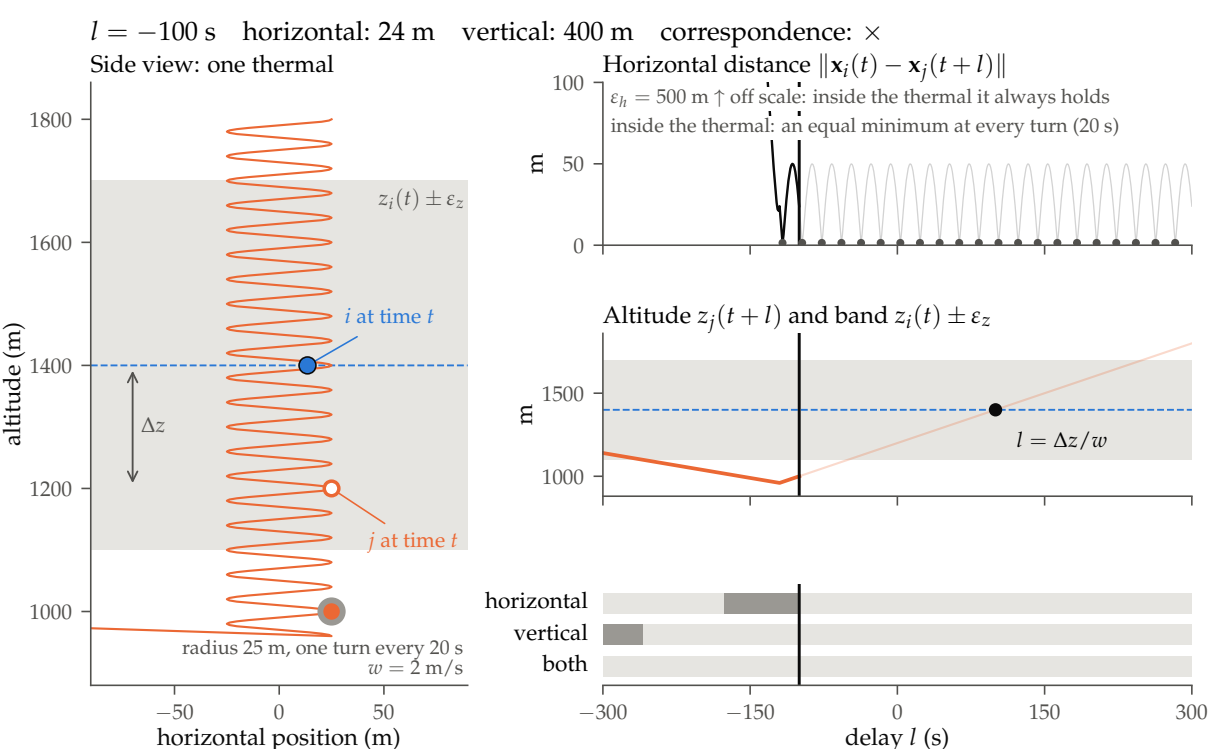

$l = -150$ s horizontal: 259 m vertical: 410 m correspondence: ×
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
j is 245 m away
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

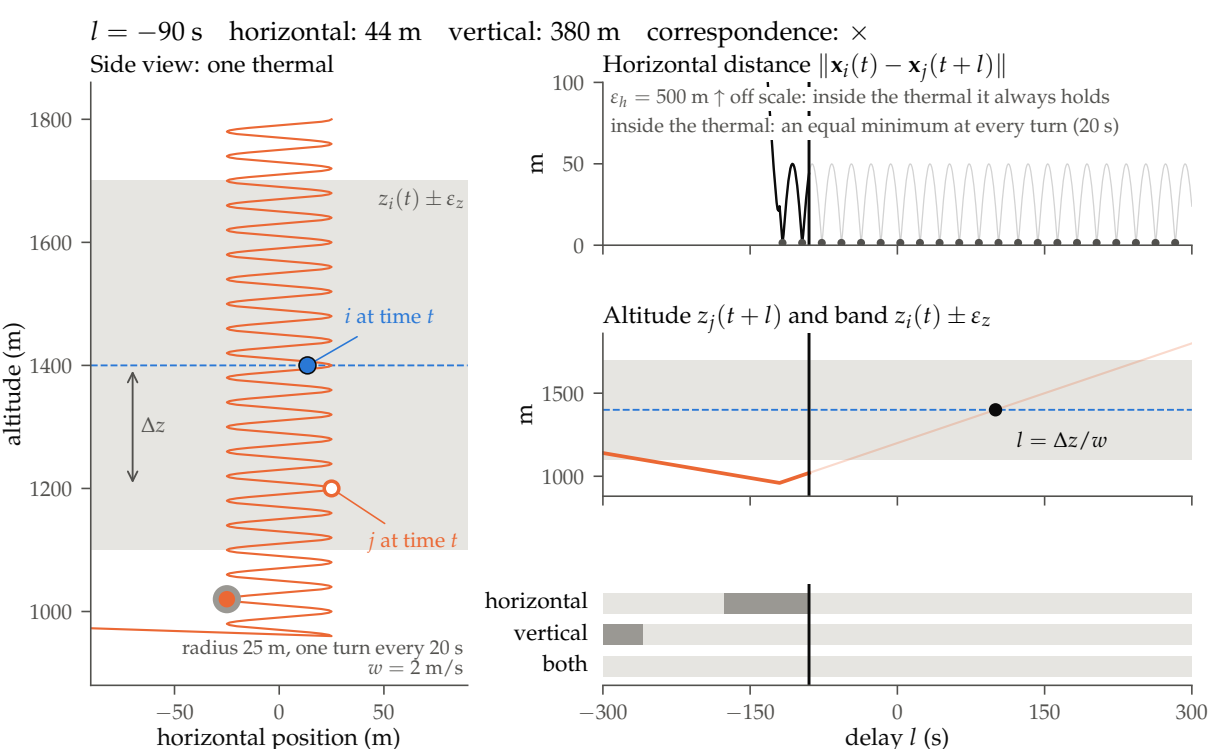

$l = -140$ s horizontal: 170 m vertical: 420 m correspondence: ×
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
j is 155 m away
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

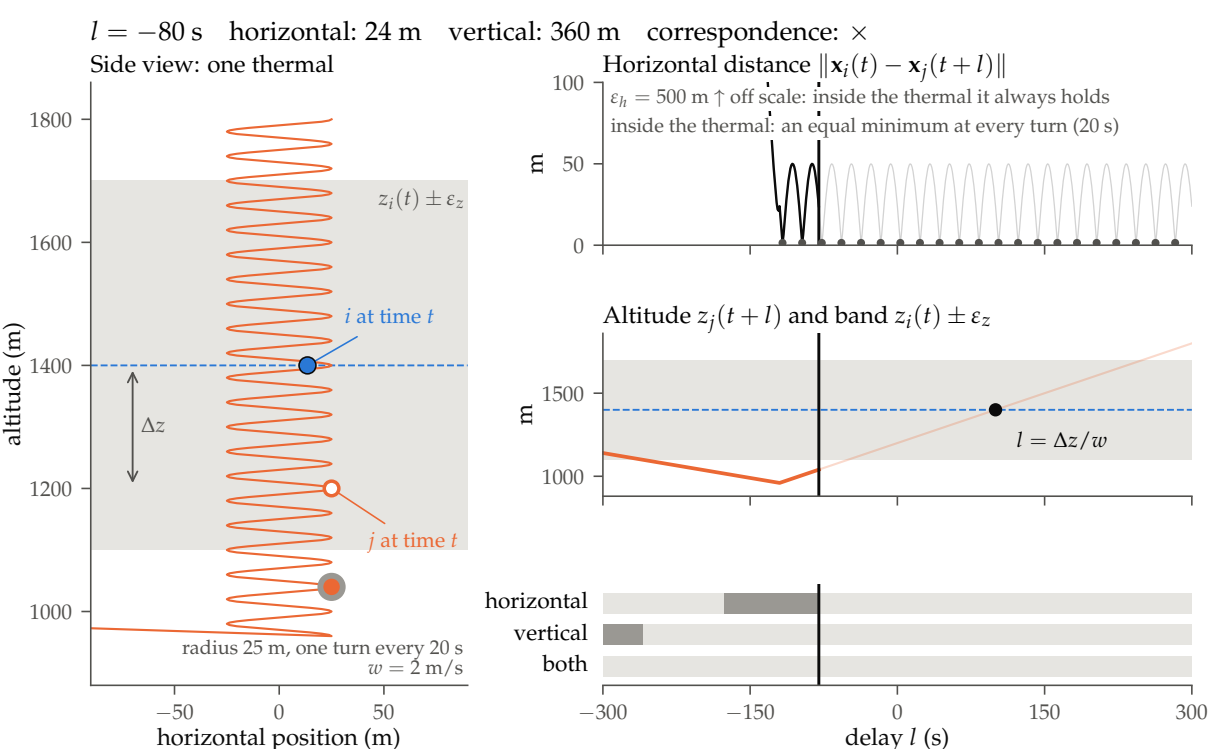

$l = -130$ s horizontal: 81 m vertical: 430 m correspondence: ×
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

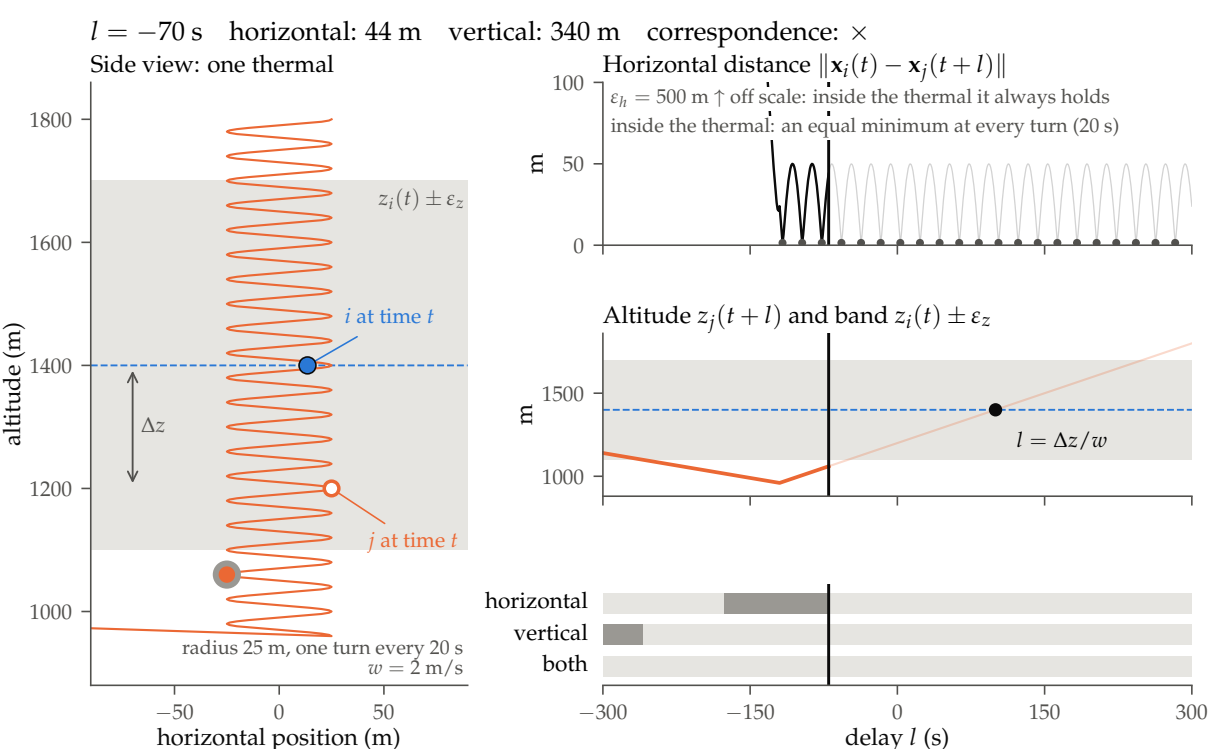

$l = -120$ s horizontal: 24 m vertical: 440 m correspondence: ×
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

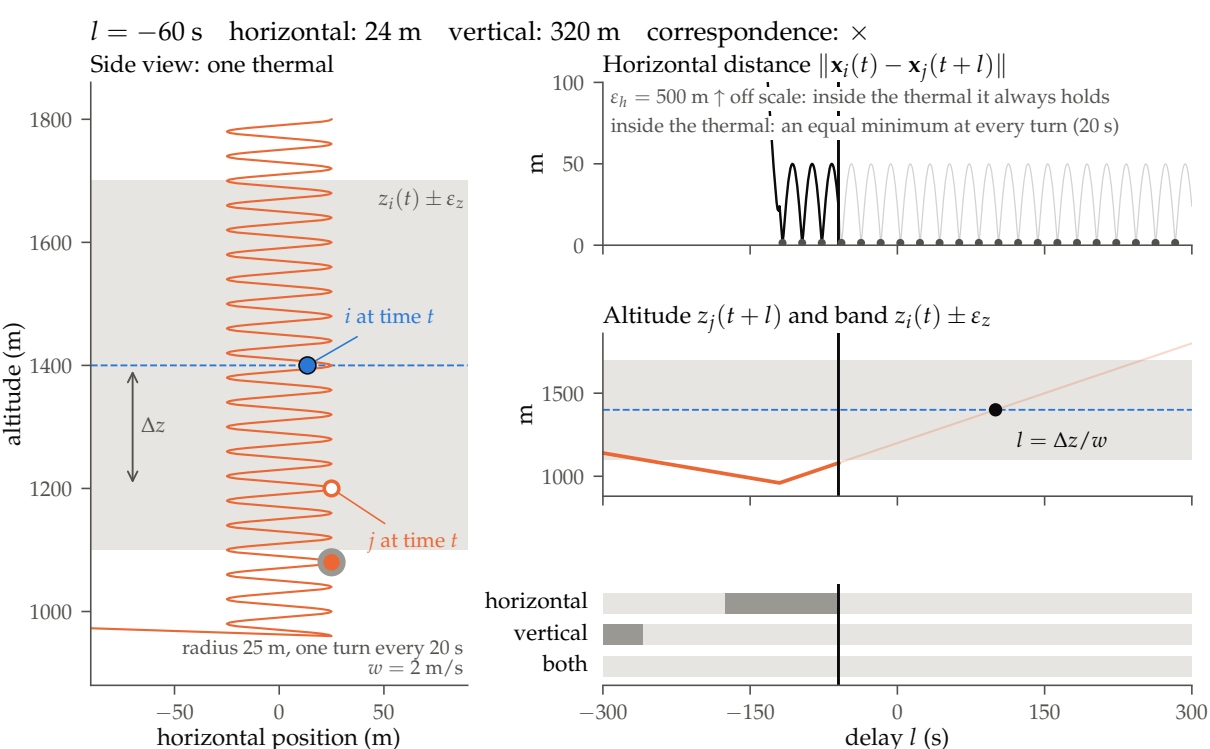

$l = -110$ s horizontal: 44 m vertical: 420 m correspondence: ×
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

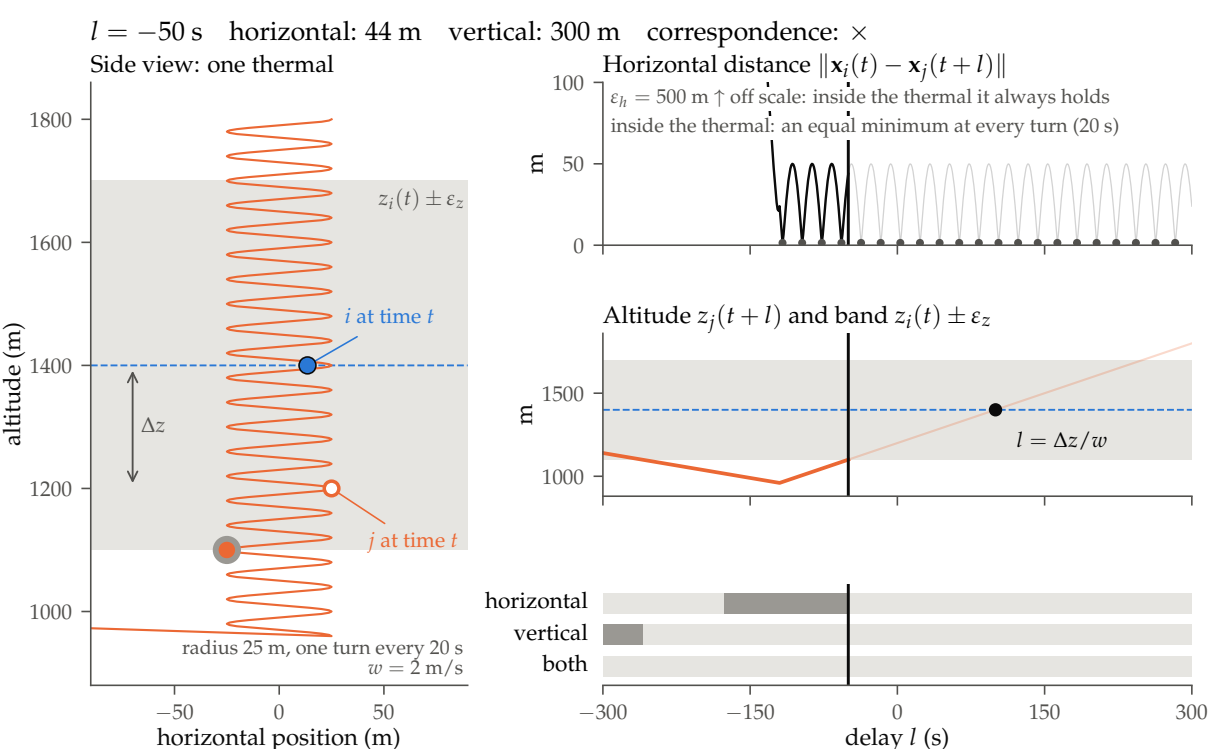

$l = -100$ s horizontal: 24 m vertical: 400 m correspondence: ×
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

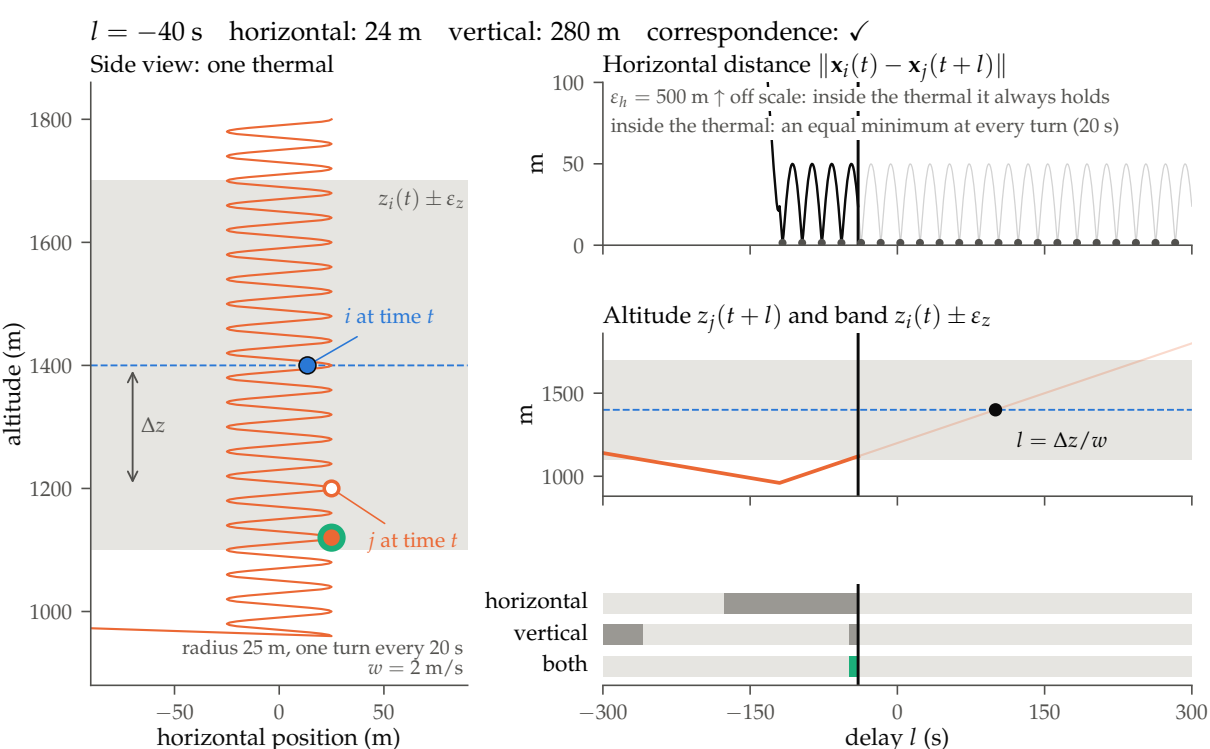

$l = -90$ s horizontal: 44 m vertical: 380 m correspondence: ×
Side view: one thermal
altitude (m)
1800
1600
1400
1200
1000
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

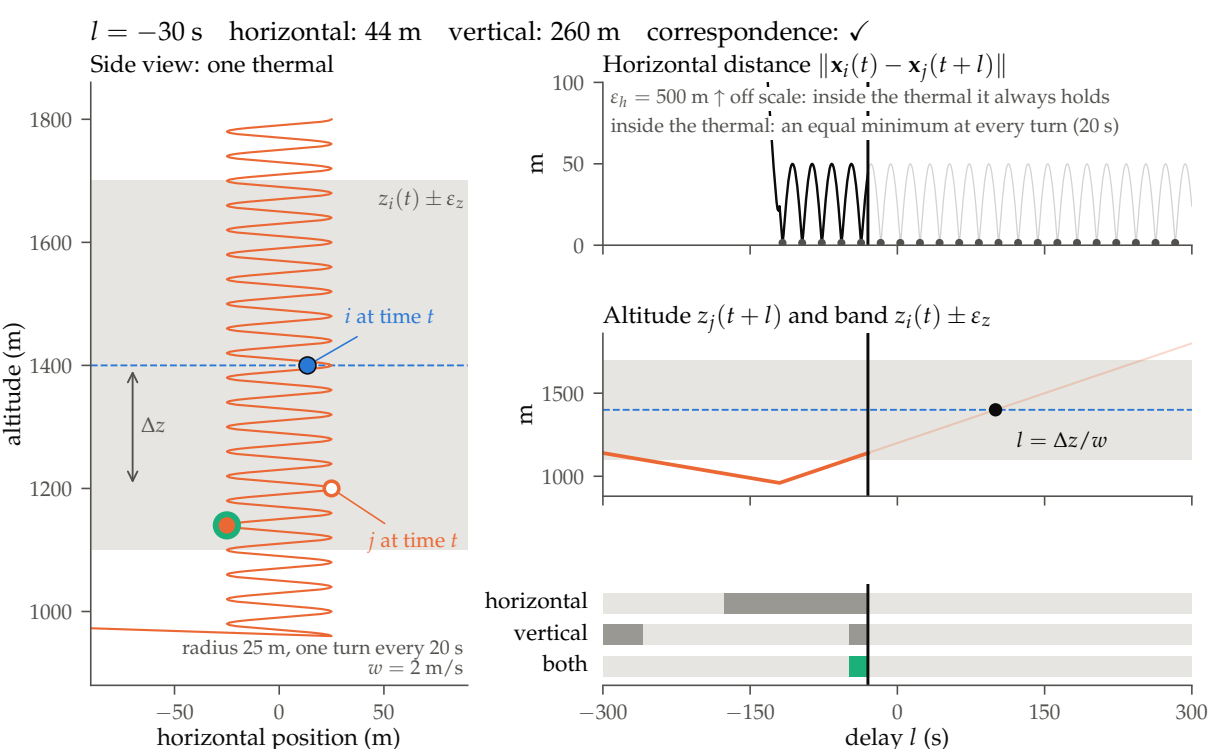

$l = -80$ s horizontal: 24 m vertical: 360 m correspondence: ×
Side view: one thermal
altitude (m)
1800
1600
1400
1200
1000
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

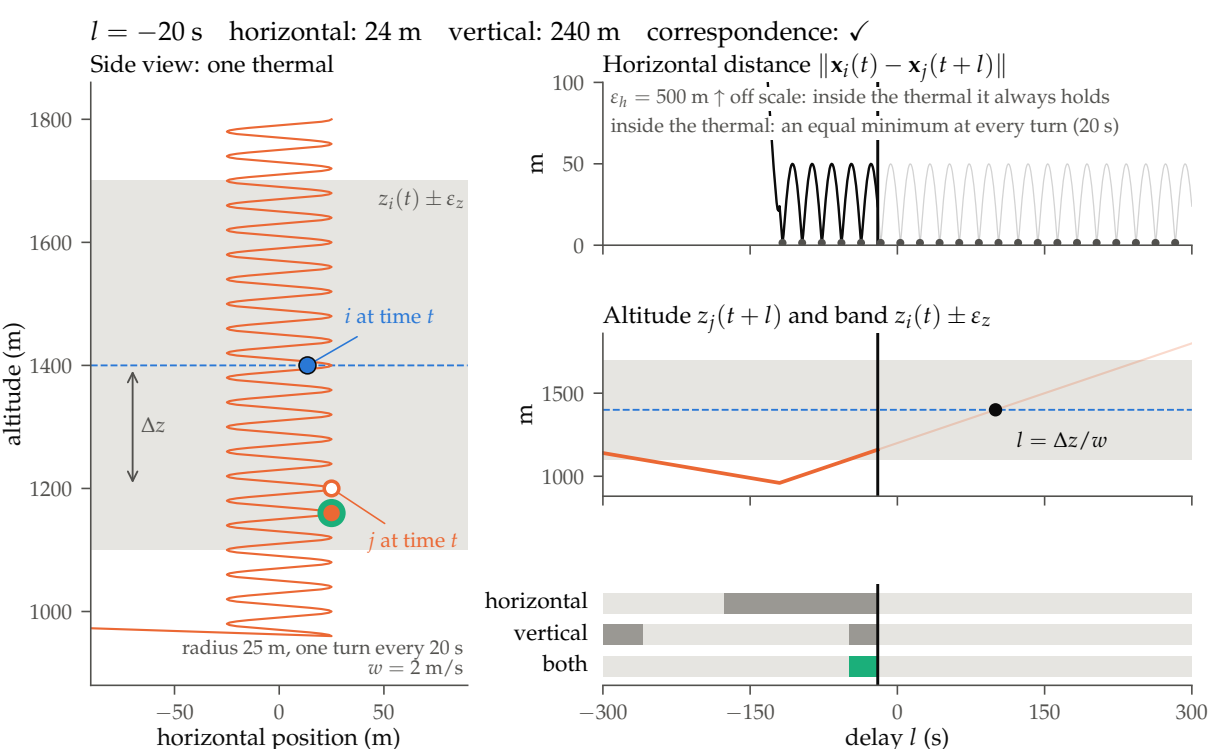

$l = -70$ s horizontal: 44 m vertical: 340 m correspondence: ×
Side view: one thermal
altitude (m)
1800
1600
1400
1200
1000
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

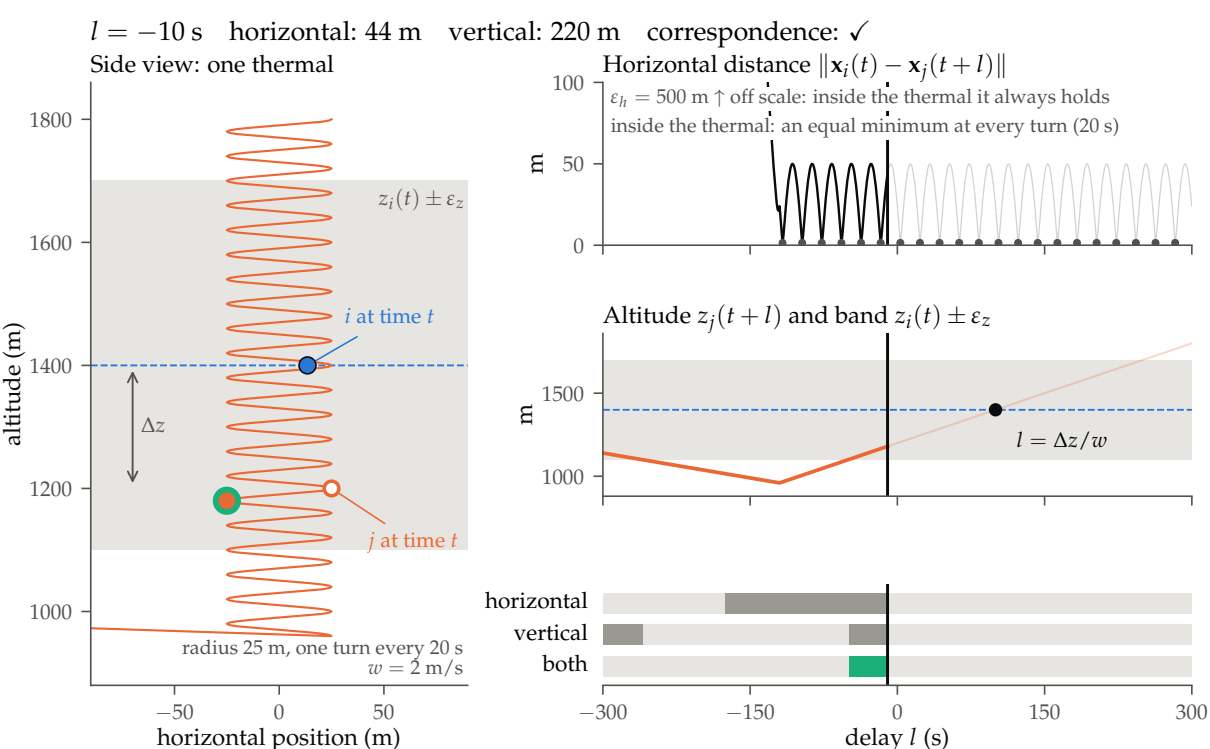

$l = -60$ s horizontal: 24 m vertical: 320 m correspondence: ×
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

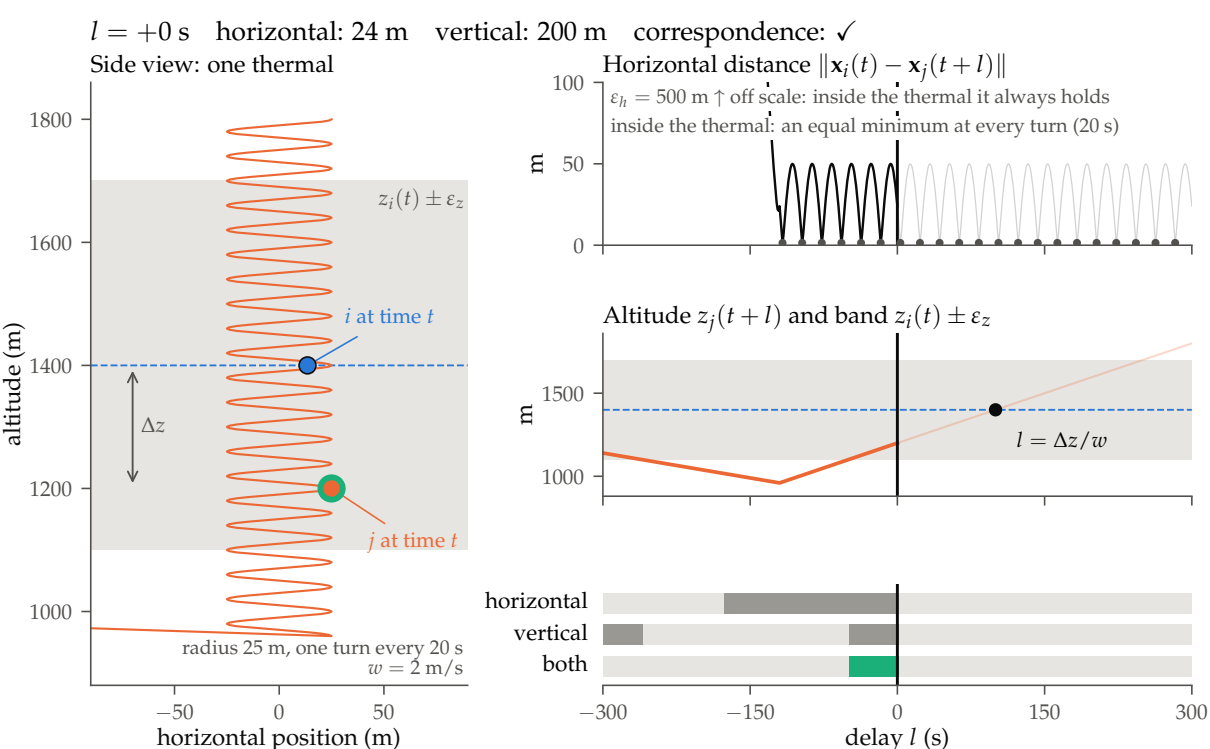

$l = -50$ s horizontal: 44 m vertical: 300 m correspondence: ×
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

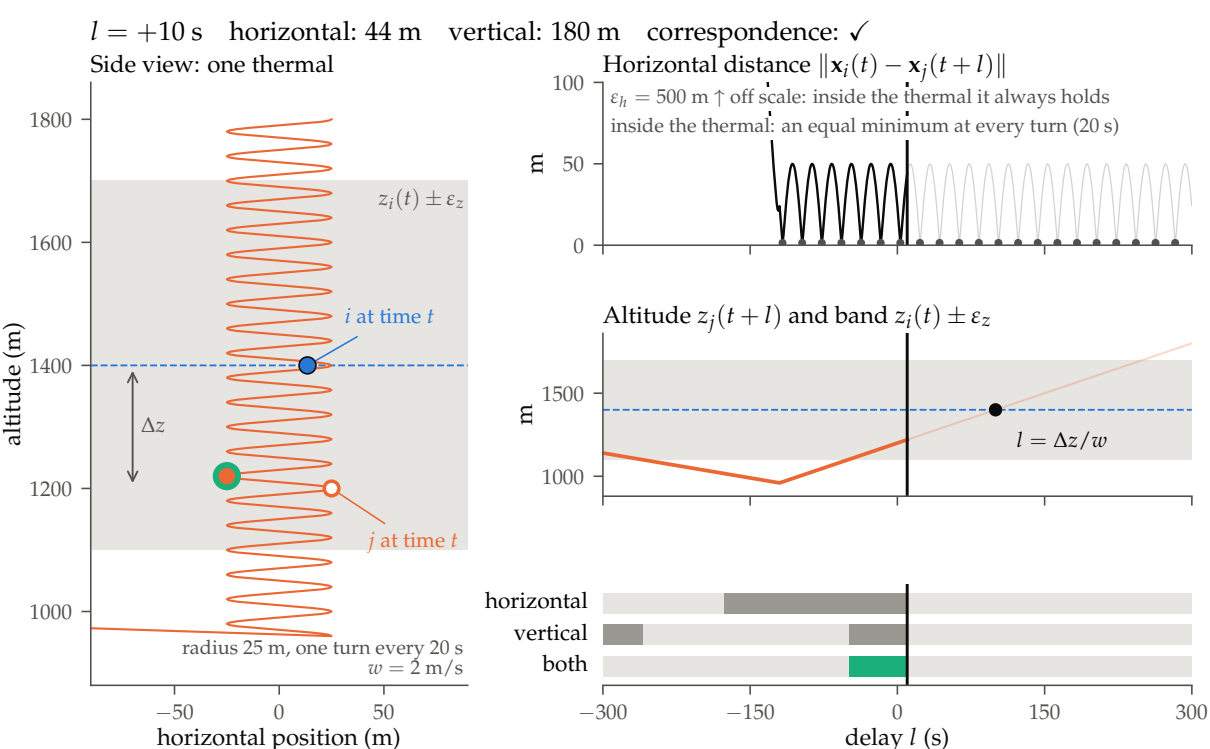

$l = -40$ s horizontal: 24 m vertical: 280 m correspondence: ✓
Side view: one thermal
altitude (m)
1800
1600
1400
1200
1000
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

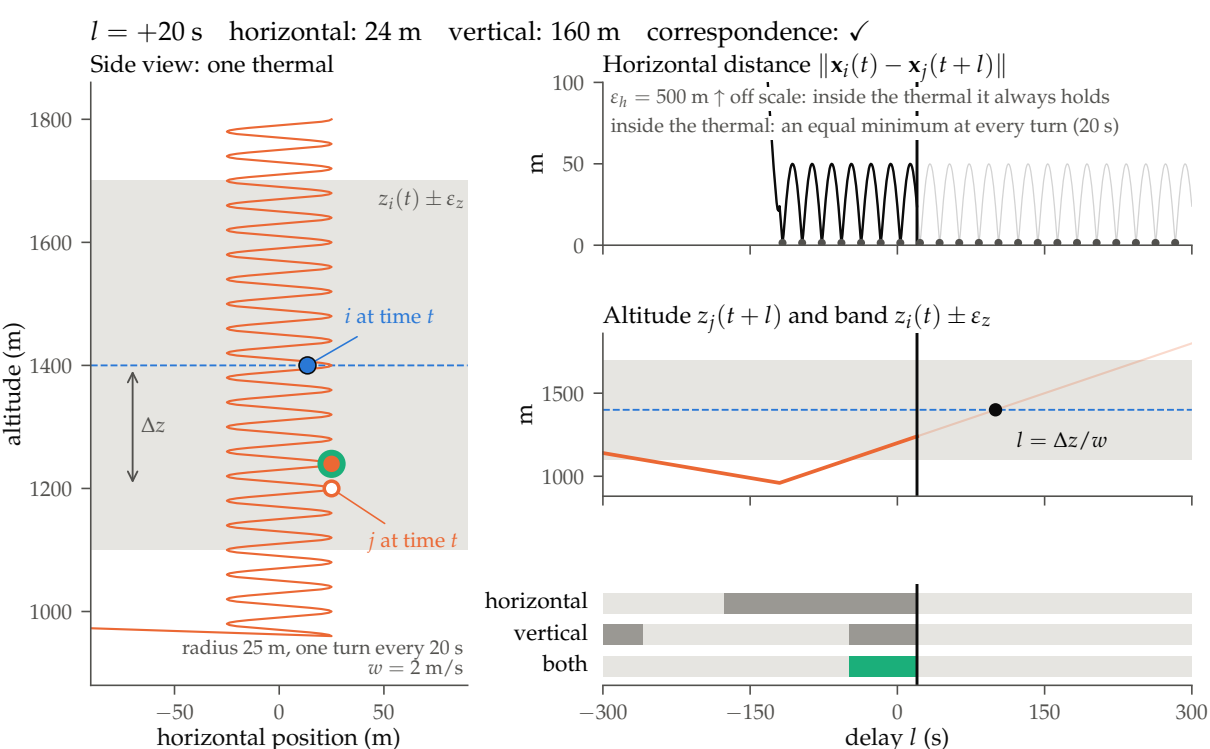

$l = -30$ s horizontal: 44 m vertical: 260 m correspondence: ✓
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

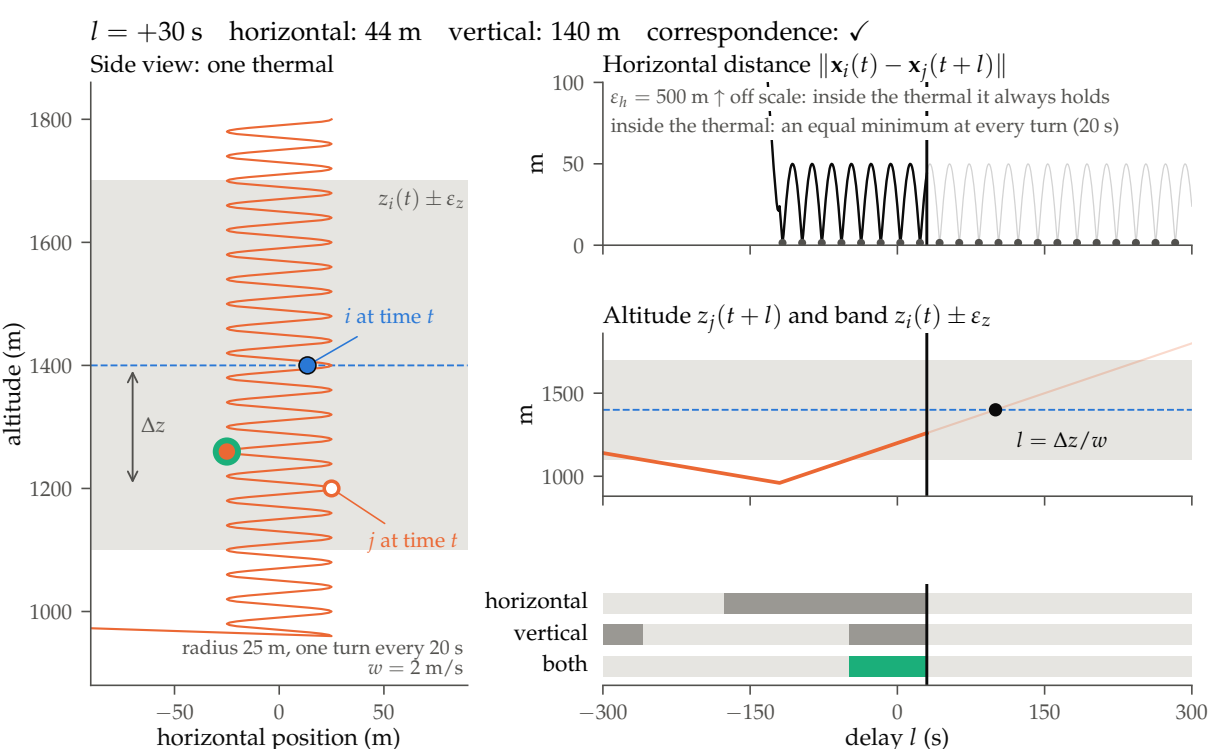

$l = -20$ s horizontal: 24 m vertical: 240 m correspondence: ✓
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

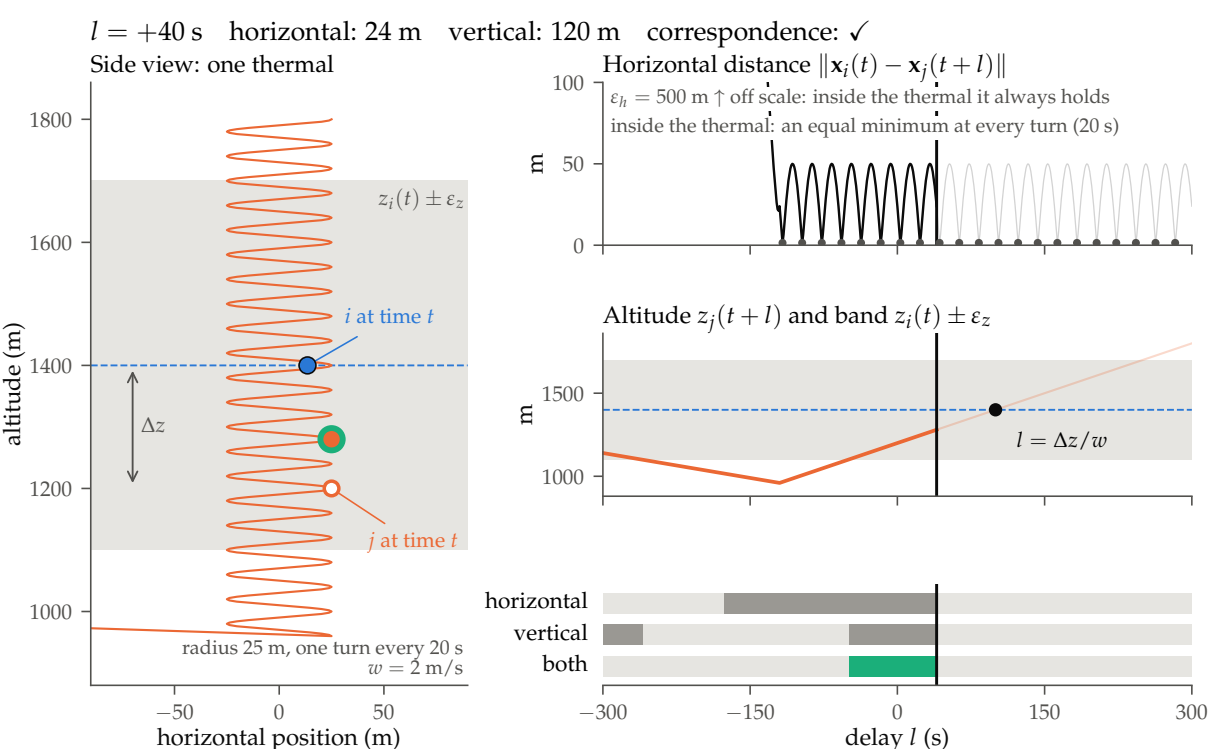

$l = -10$ s horizontal: 44 m vertical: 220 m correspondence: ✓
Side view: one thermal
altitude (m)
1800
1600
1400
1200
1000
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
m
100
50
0
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
m
1500
1000
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

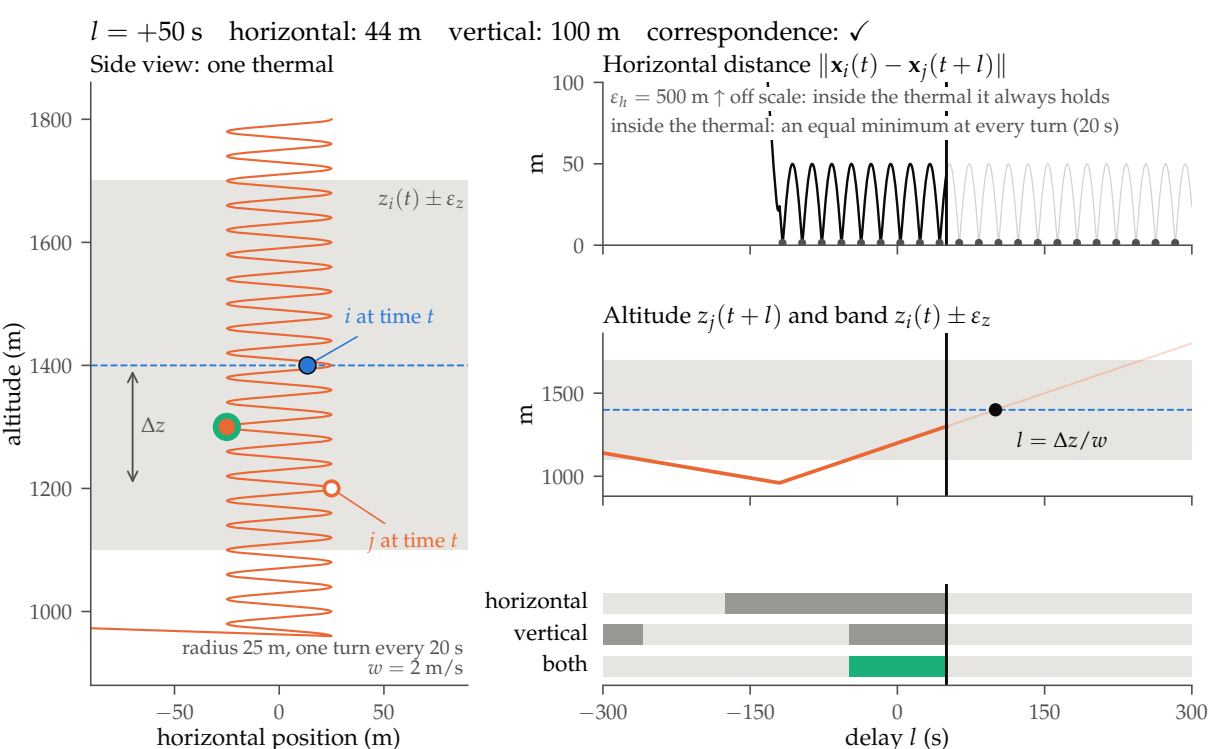

$l = +0$ s horizontal: 24 m vertical: 200 m correspondence: ✓
Side view: one thermal
altitude (m)
1800
1600
1400
1200
1000
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

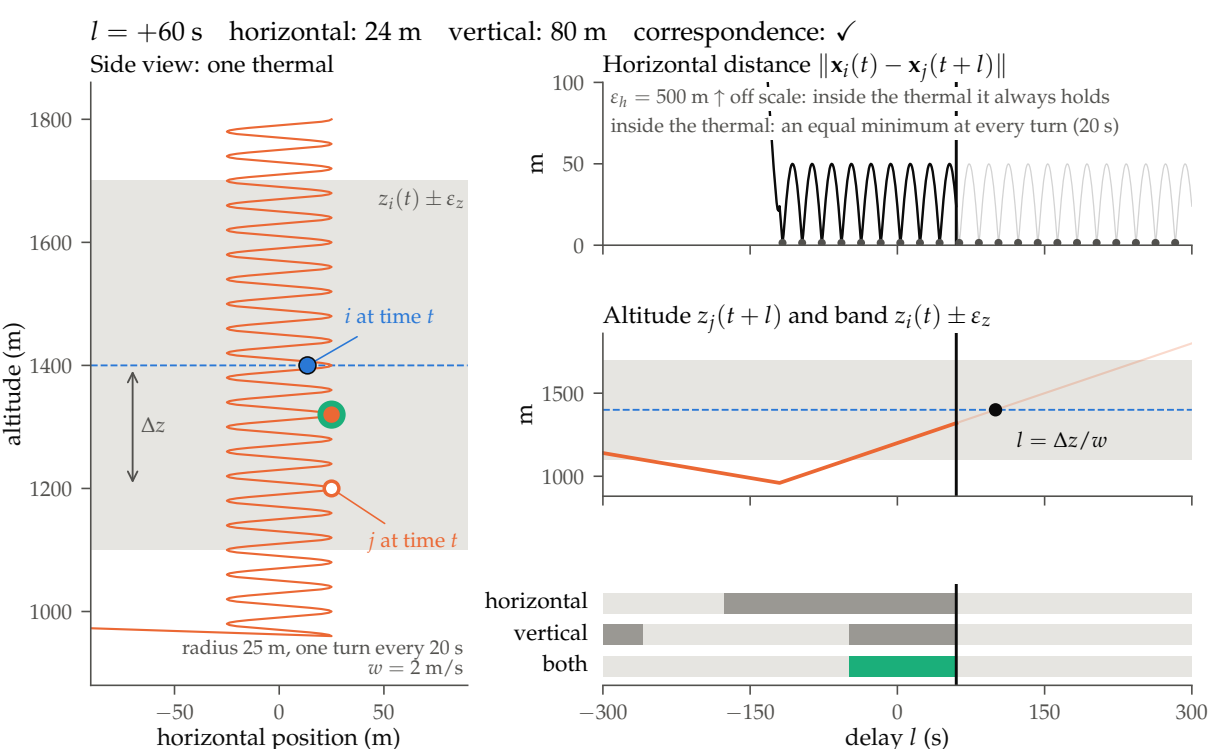

$l = +10$ s horizontal: 44 m vertical: 180 m correspondence: ✓
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

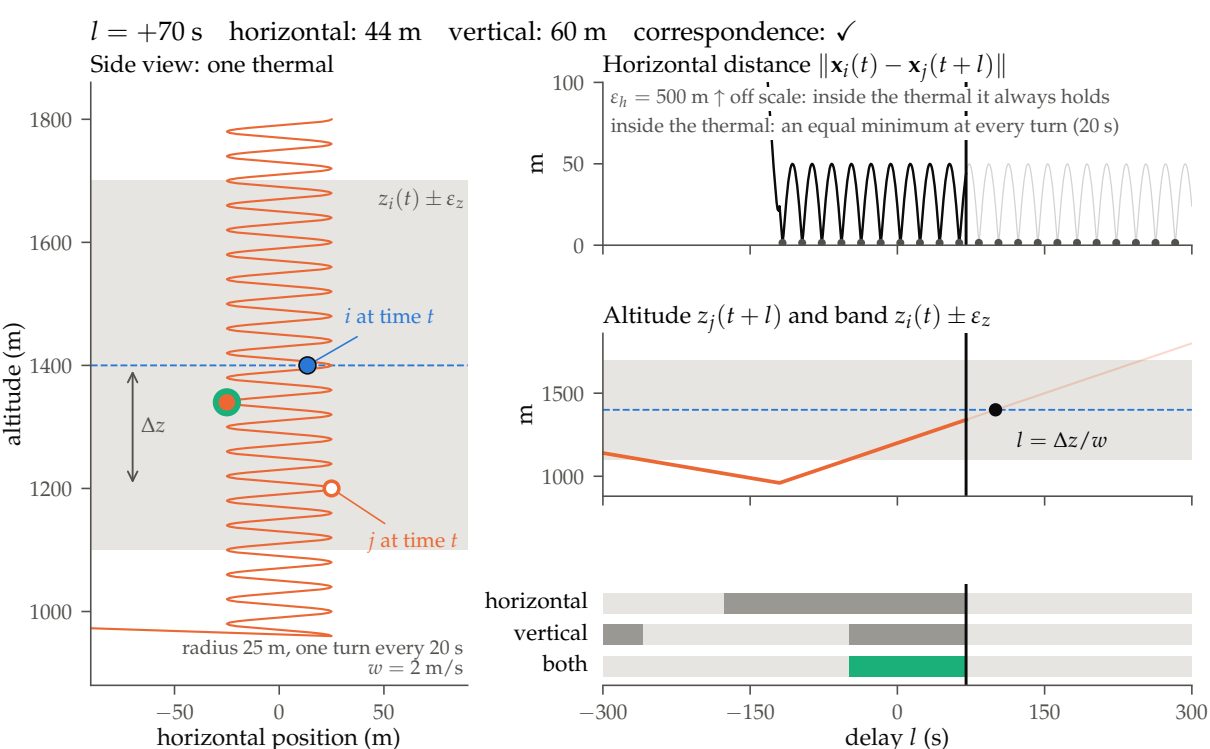

$l = +20$ s horizontal: 24 m vertical: 160 m correspondence: ✓
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

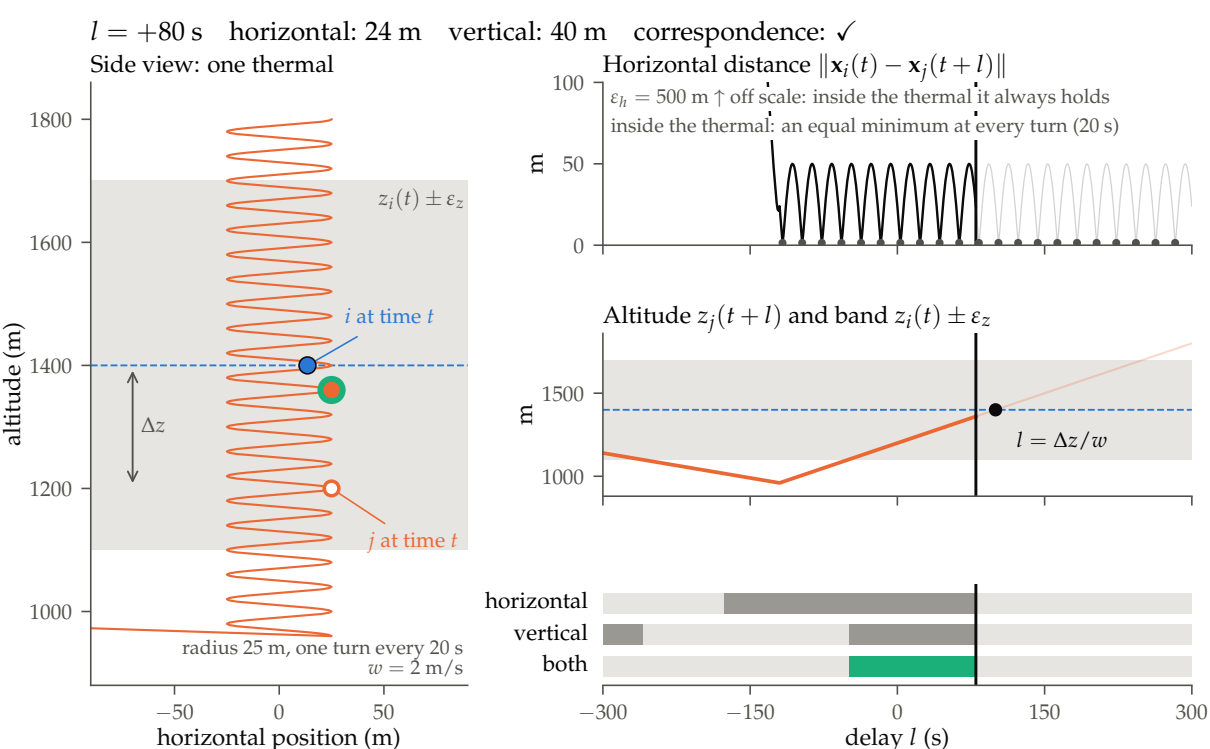

$l = +30$ s horizontal: 44 m vertical: 140 m correspondence: ✓
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

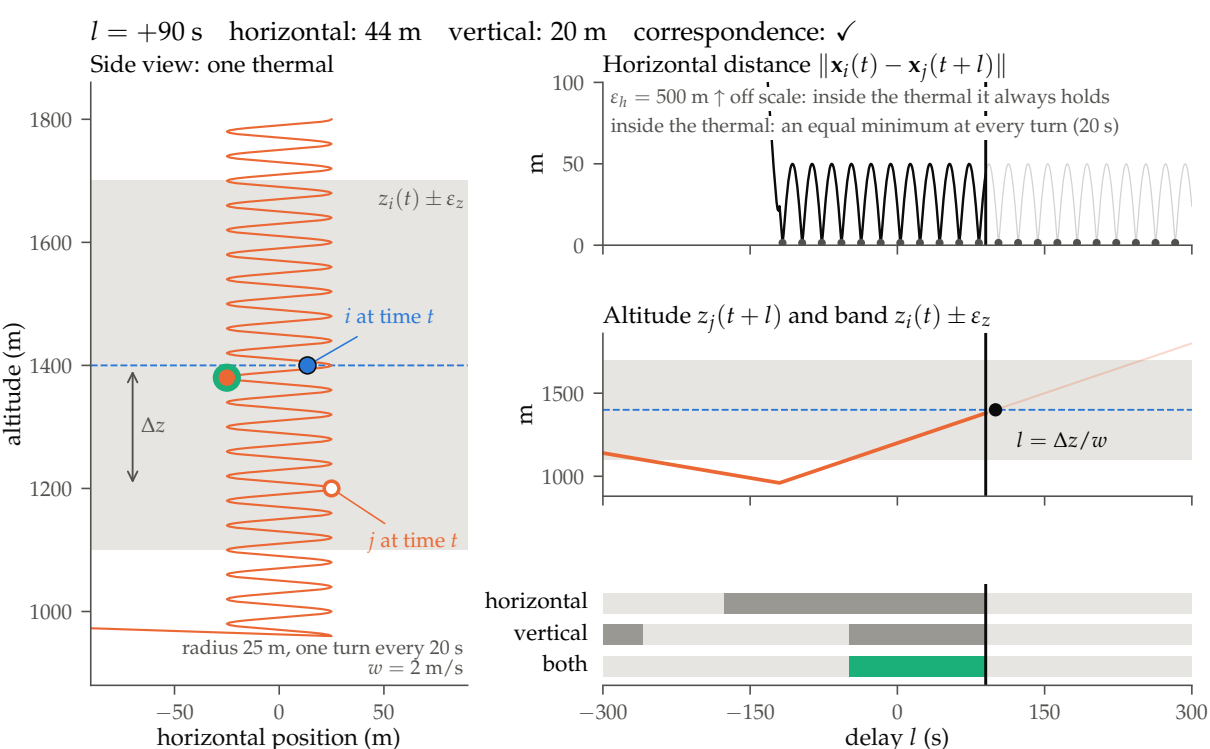

$l = +40$ s horizontal: 24 m vertical: 120 m correspondence: ✓
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

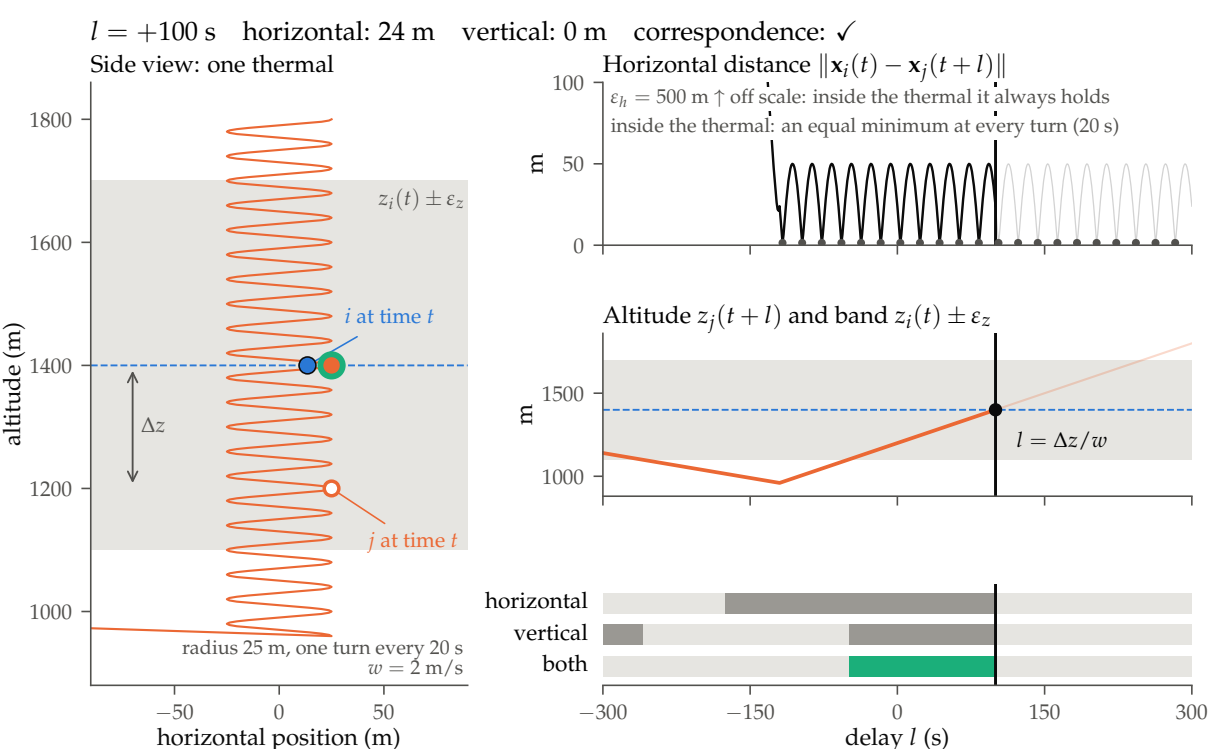

$l = +50$ s horizontal: 44 m vertical: 100 m correspondence: ✓
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

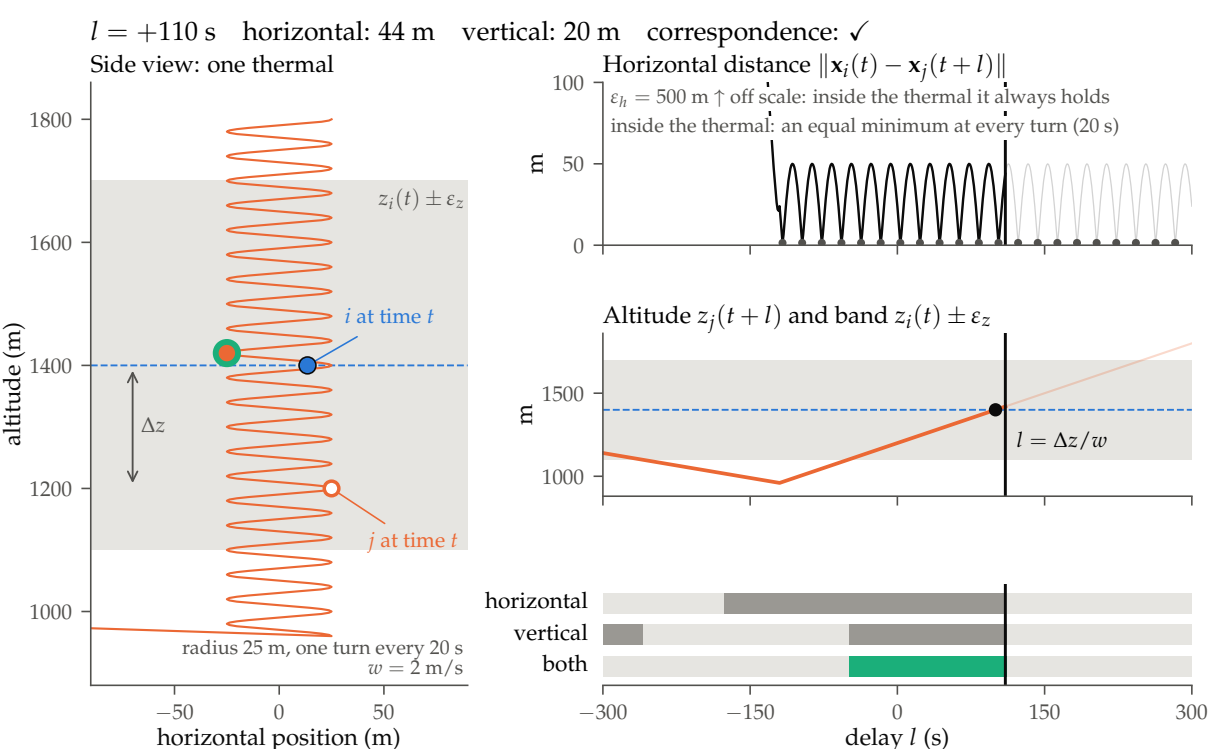

$l = +60$ s horizontal: 24 m vertical: 80 m correspondence: ✓
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

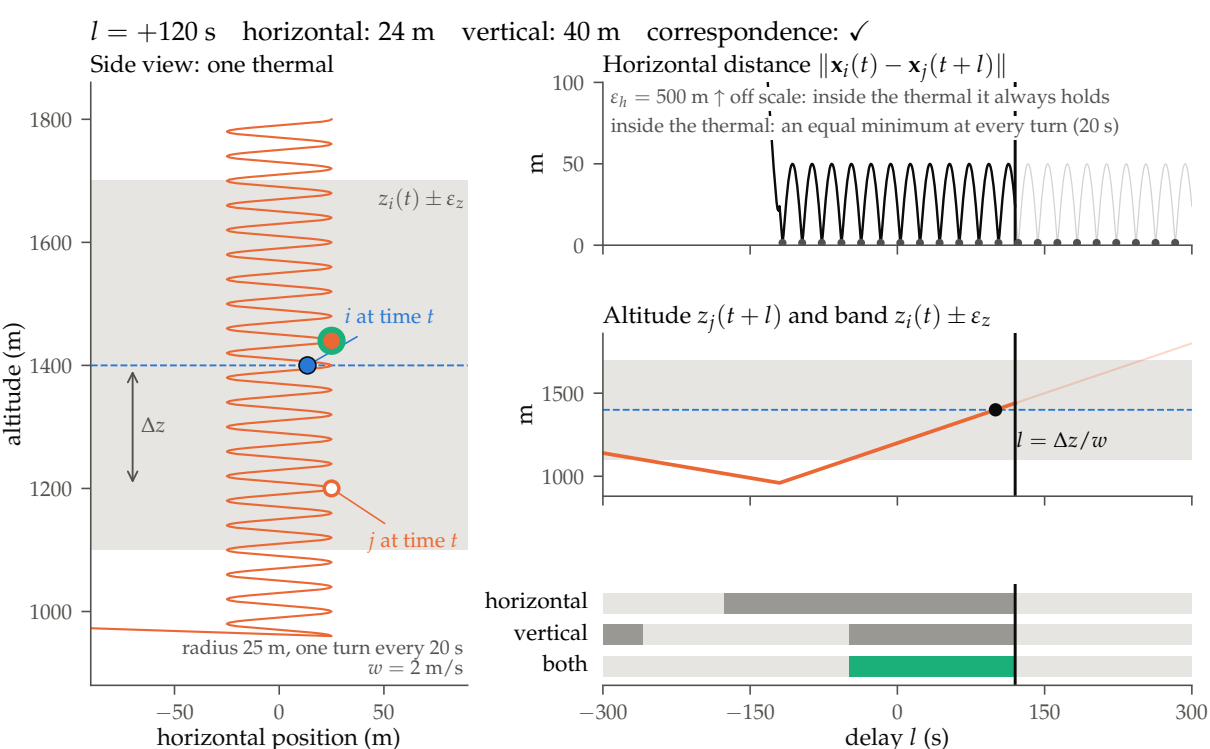

$l = +70$ s horizontal: 44 m vertical: 60 m correspondence: ✓
Side view: one thermal
altitude (m)
1800
1600
1400
1200
1000
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

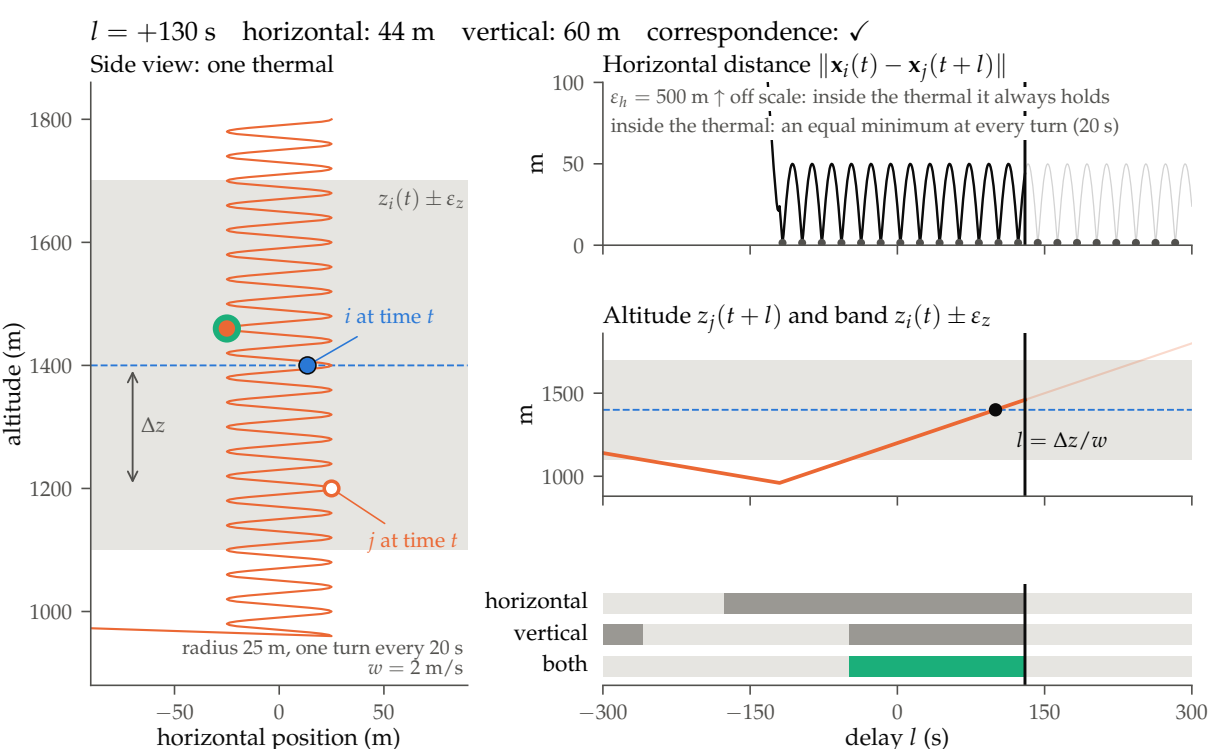

$l = +80$ s horizontal: 24 m vertical: 40 m correspondence: ✓
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

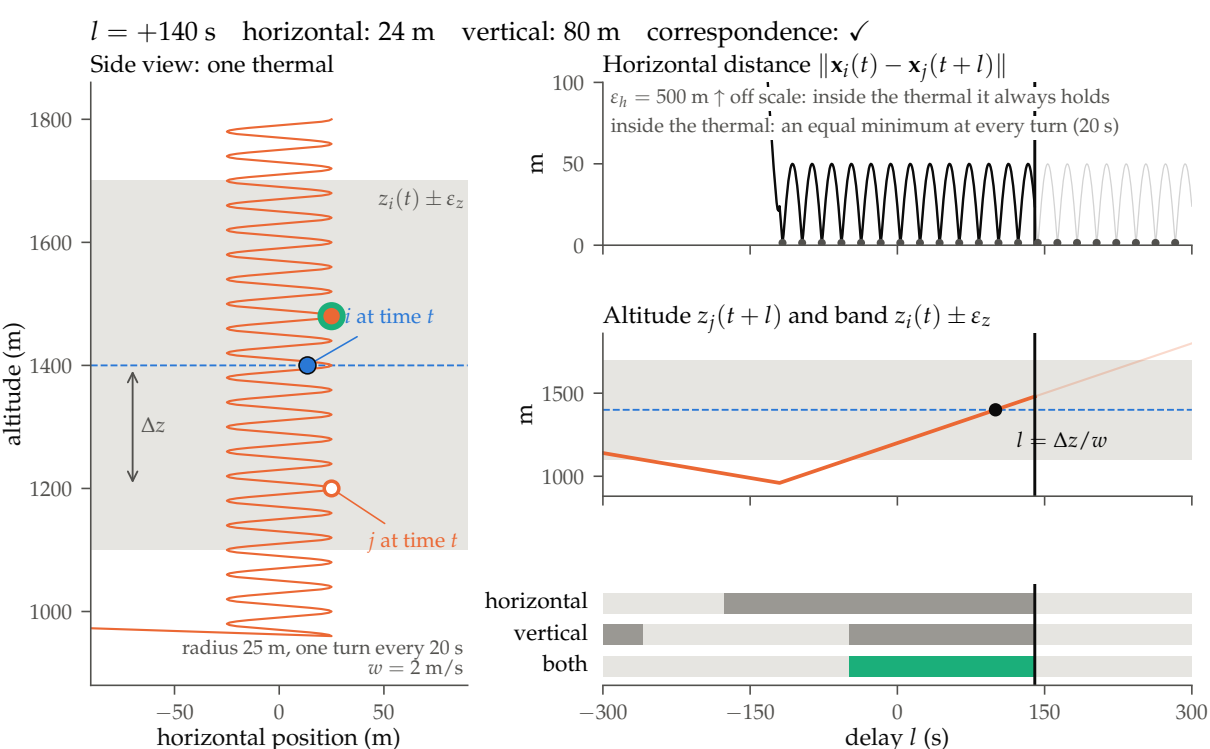

$l = +90$ s horizontal: 44 m vertical: 20 m correspondence: ✓
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

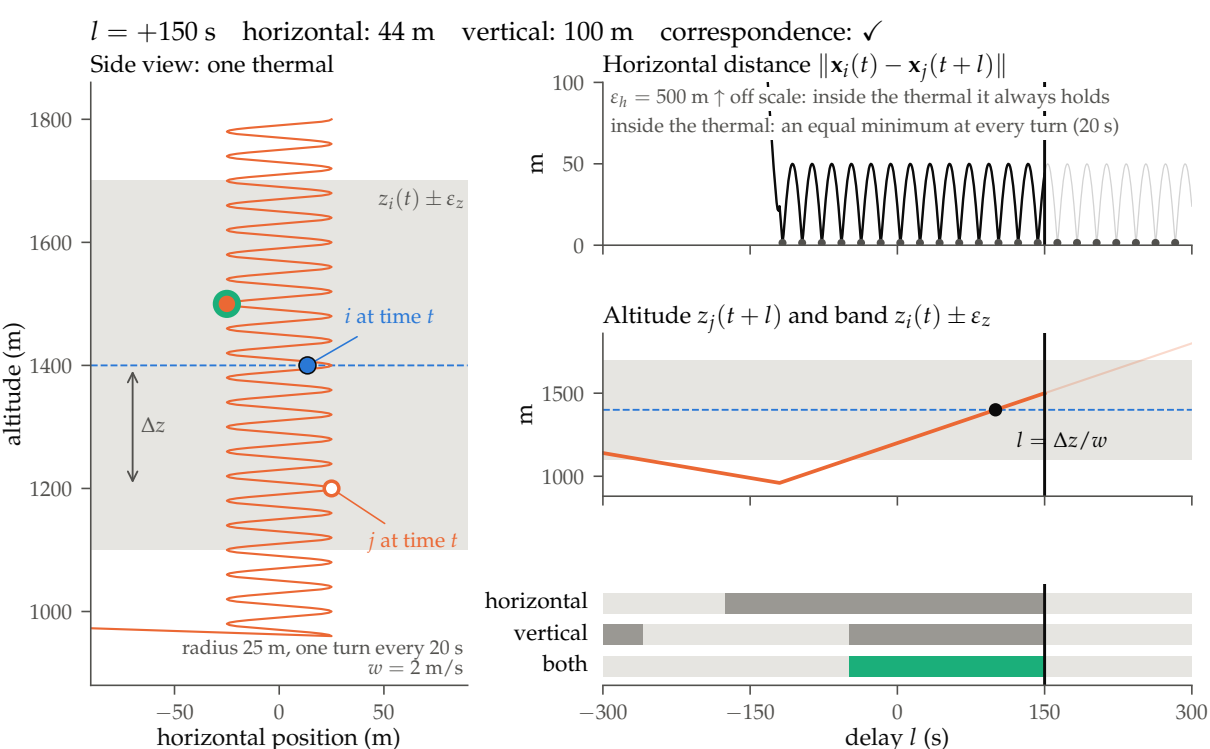

$l = +100$ s horizontal: 24 m vertical: 0 m correspondence: ✓
Side view: one thermal
altitude (m)
1800
1600
1400
1200
1000
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

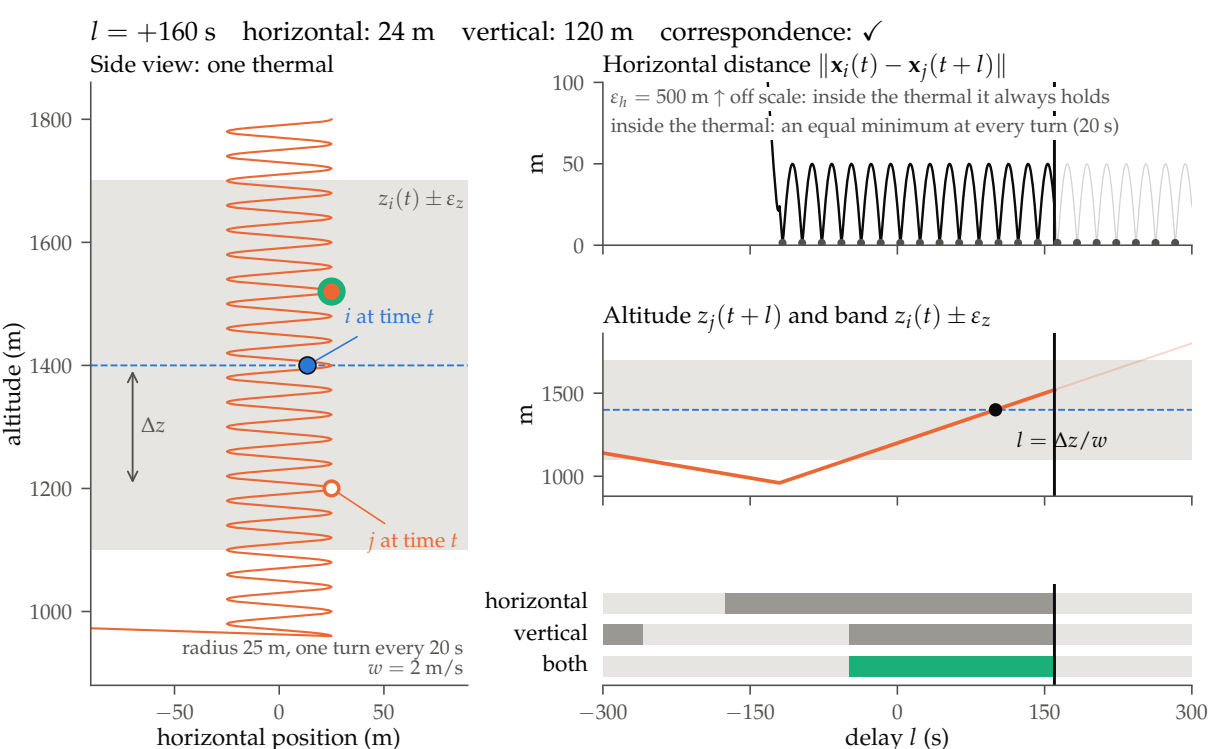

$l = +110$ s horizontal: 44 m vertical: 20 m correspondence: ✓
Side view: one thermal
altitude (m)
1800
1600
1400
1200
1000
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

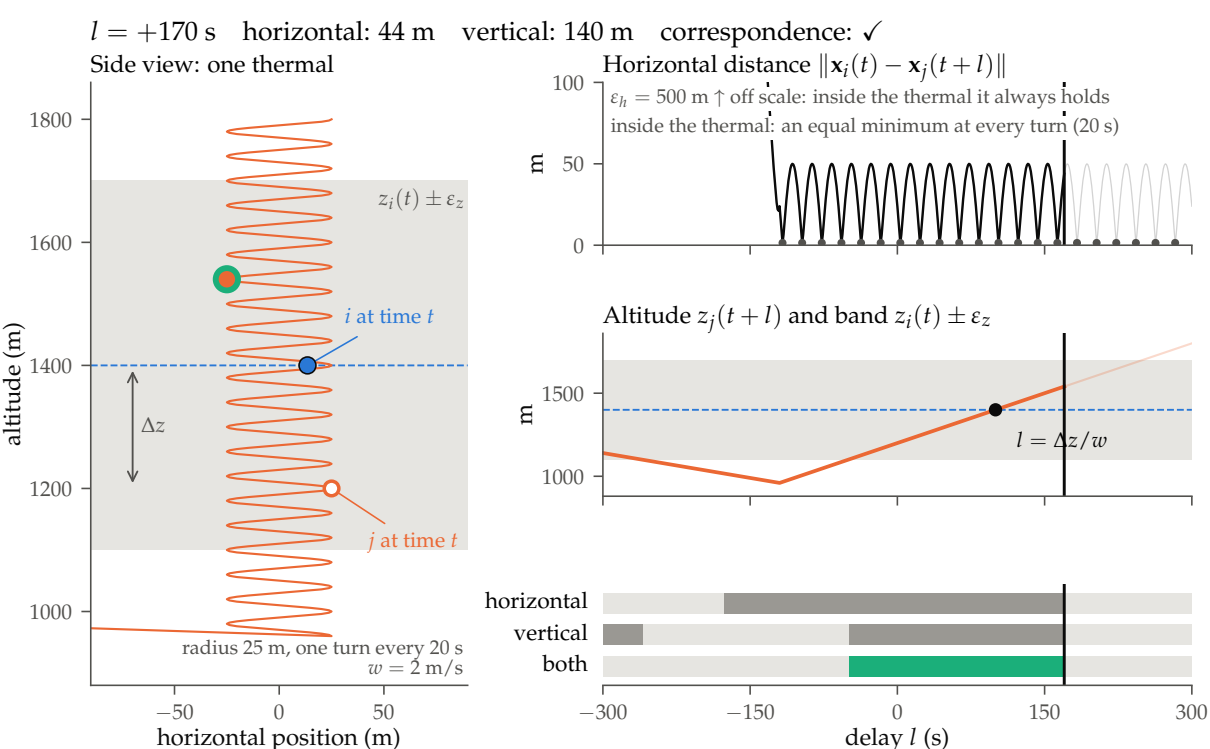

$l = +120$ s horizontal: 24 m vertical: 40 m correspondence: ✓
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

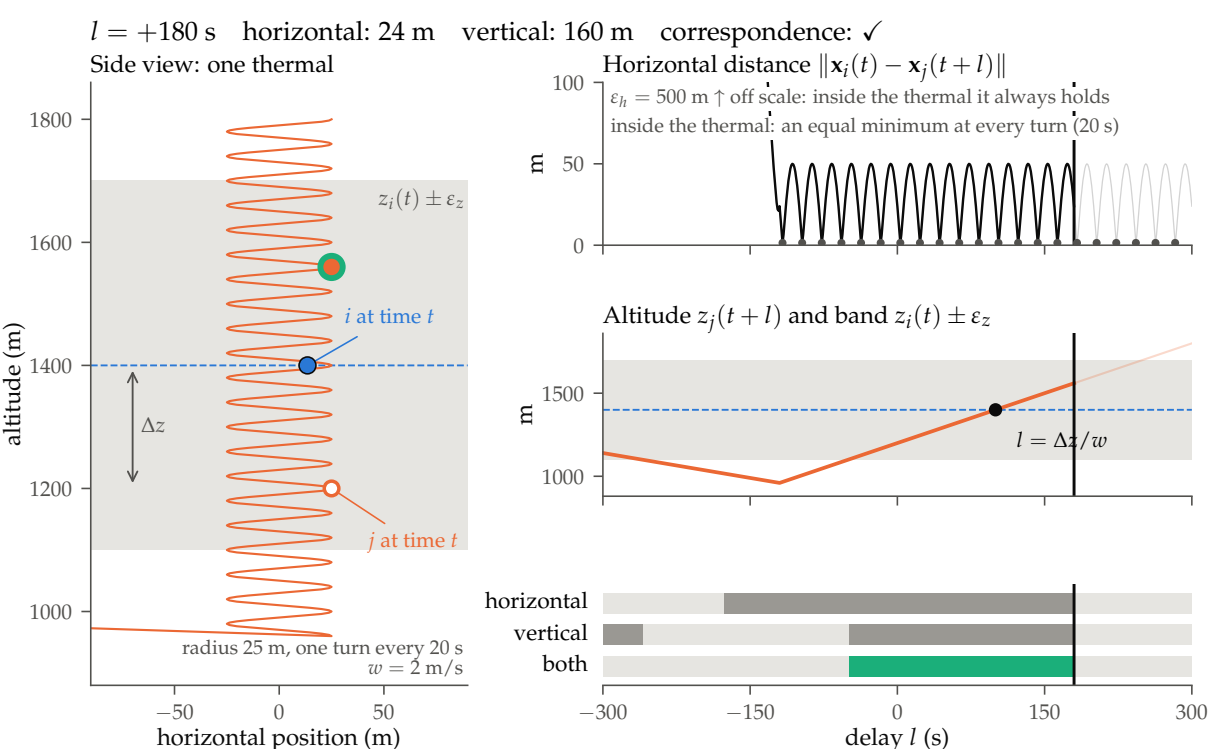

$l = +130$ s horizontal: 44 m vertical: 60 m correspondence: ✓
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

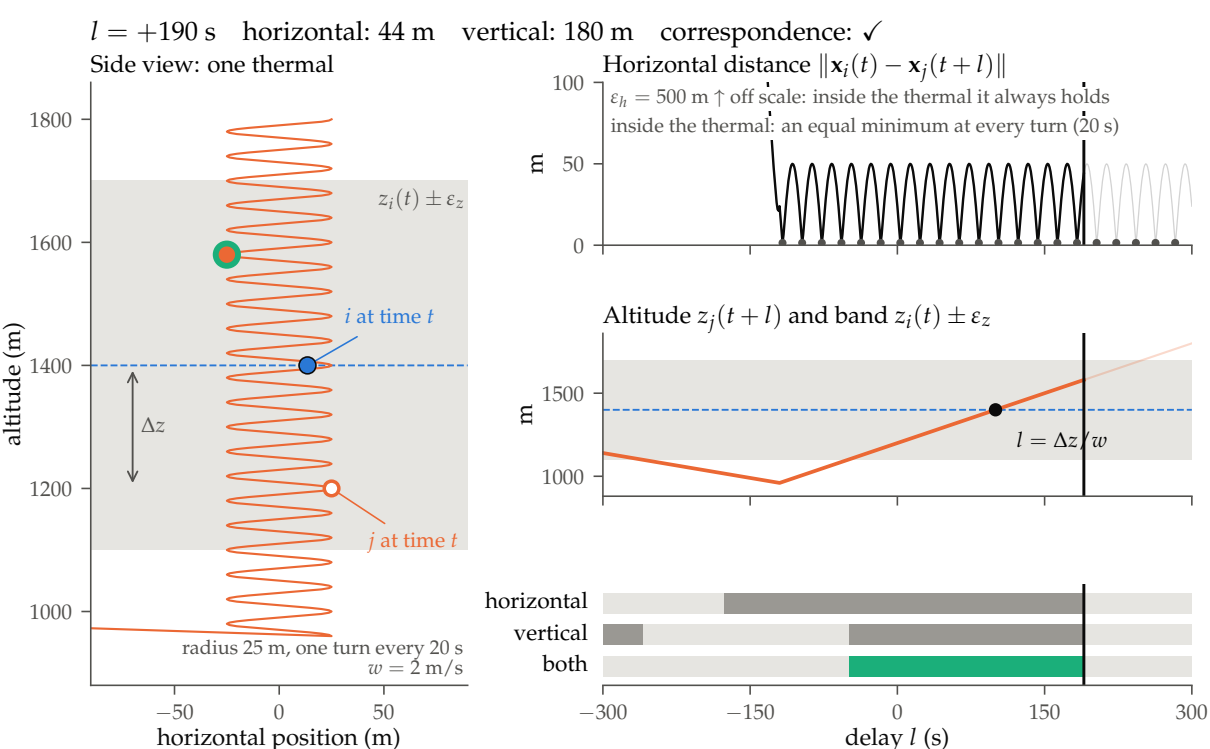

$l = +140$ s horizontal: 24 m vertical: 80 m correspondence: ✓
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

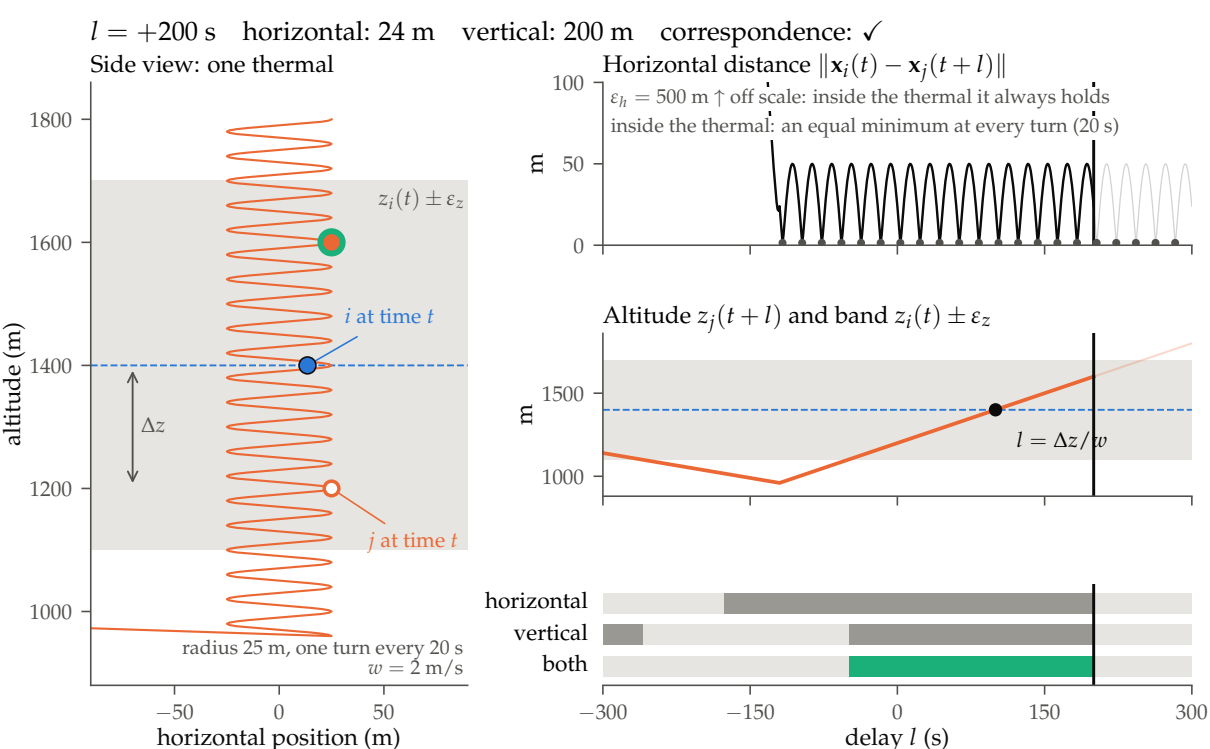

$l = +150$ s horizontal: 44 m vertical: 100 m correspondence: ✓
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

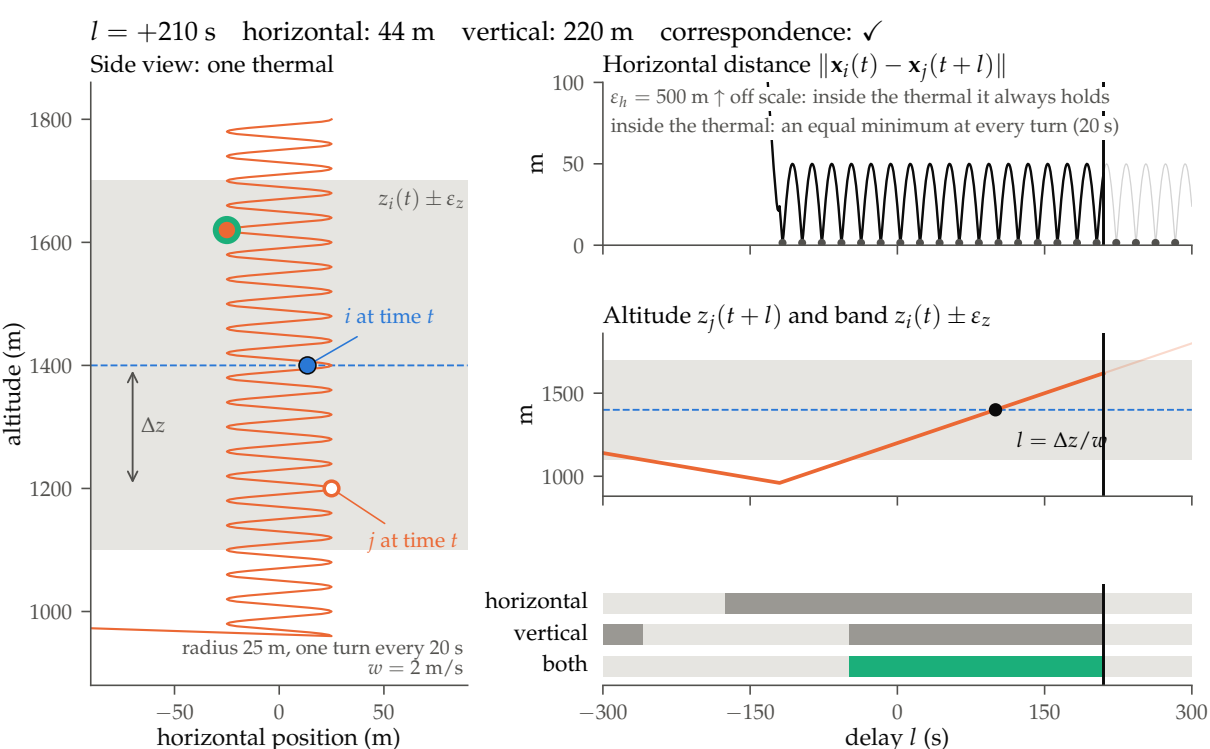

$l = +160$ s horizontal: 24 m vertical: 120 m correspondence: ✓
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

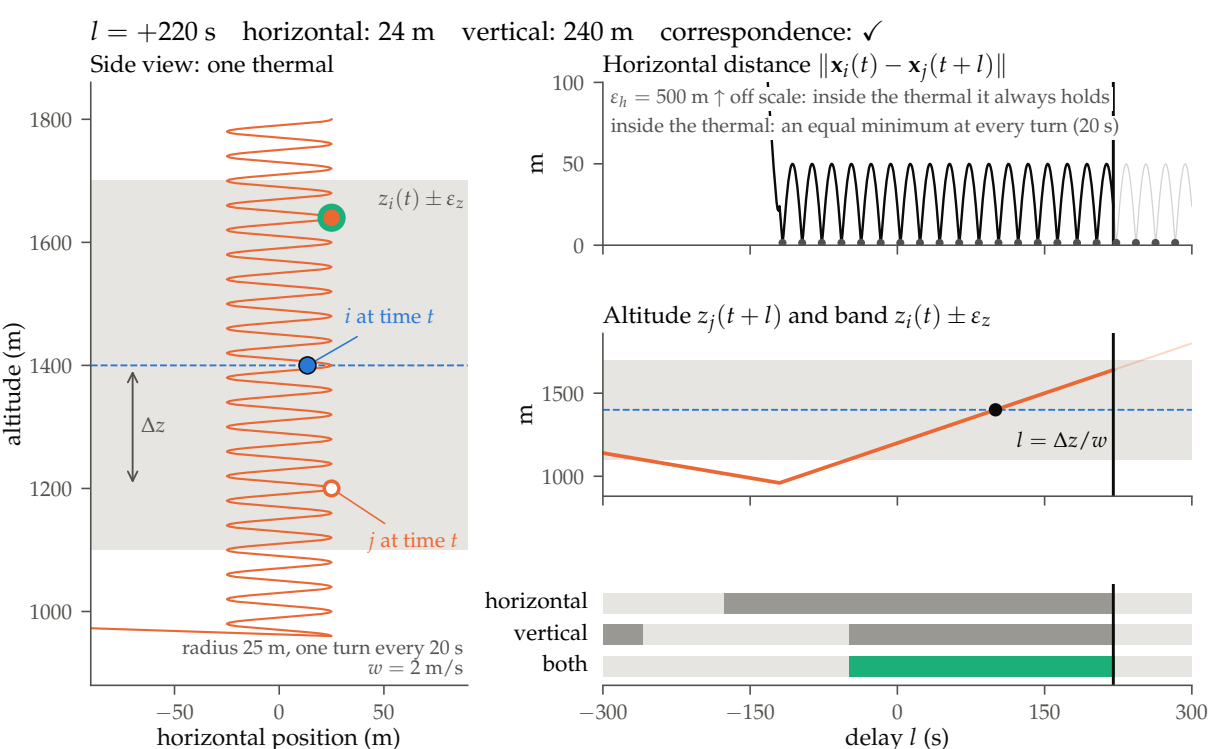

$l = +170$ s horizontal: 44 m vertical: 140 m correspondence: ✓
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

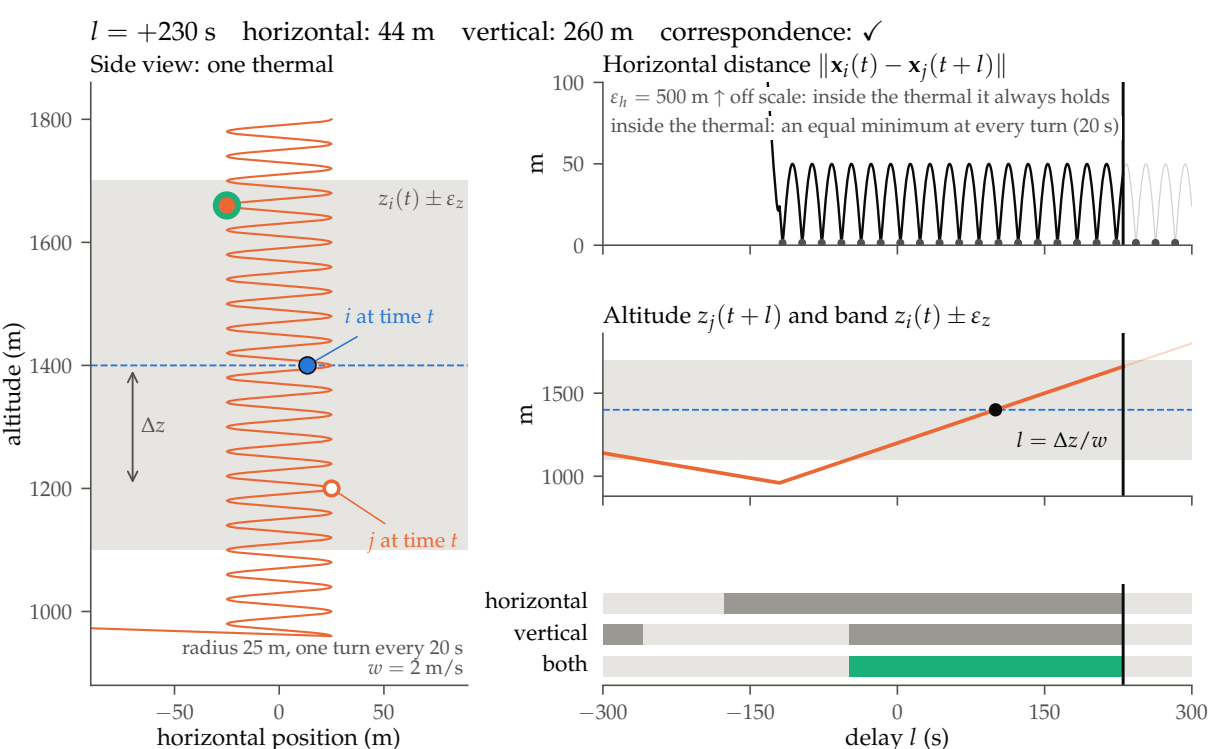

$l = +180$ s horizontal: 24 m vertical: 160 m correspondence: ✓
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

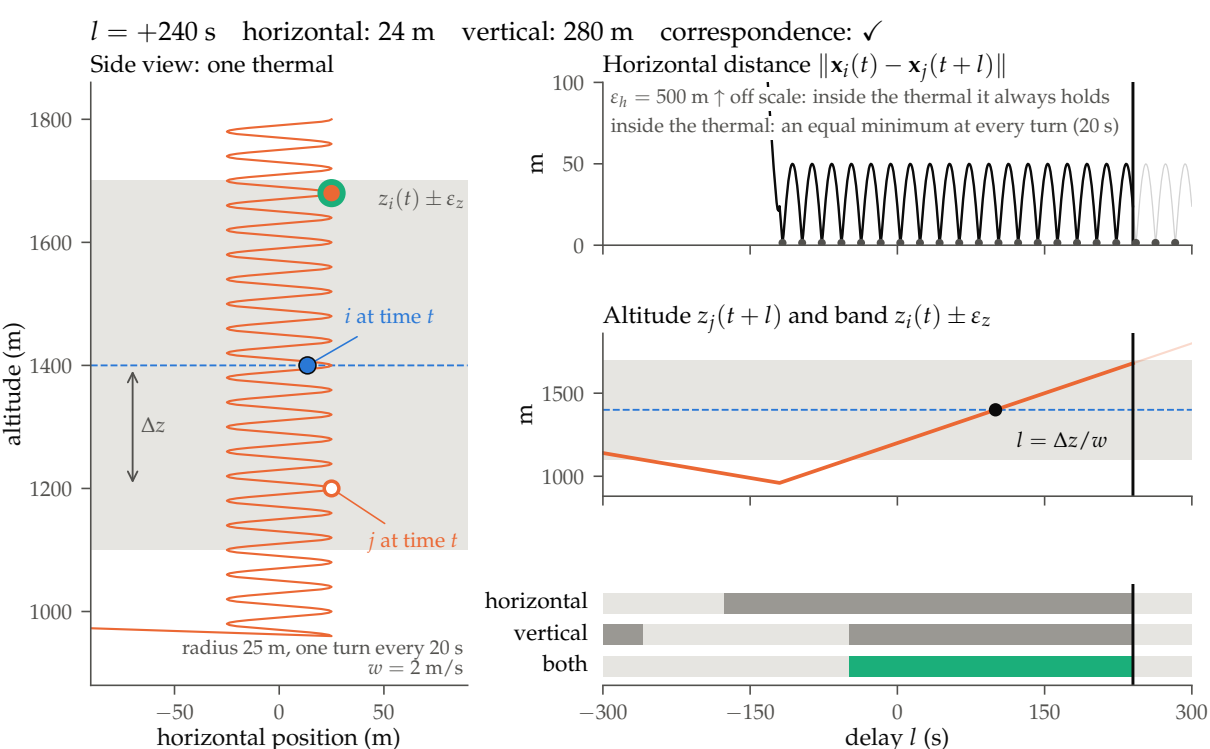

$l = +190$ s horizontal: 44 m vertical: 180 m correspondence: ✓
Side view: one thermal
altitude (m)
1800
1600
1400
1200
1000
$z_i(t) \pm \varepsilon_z$
i at time t
j at time t
Δz
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

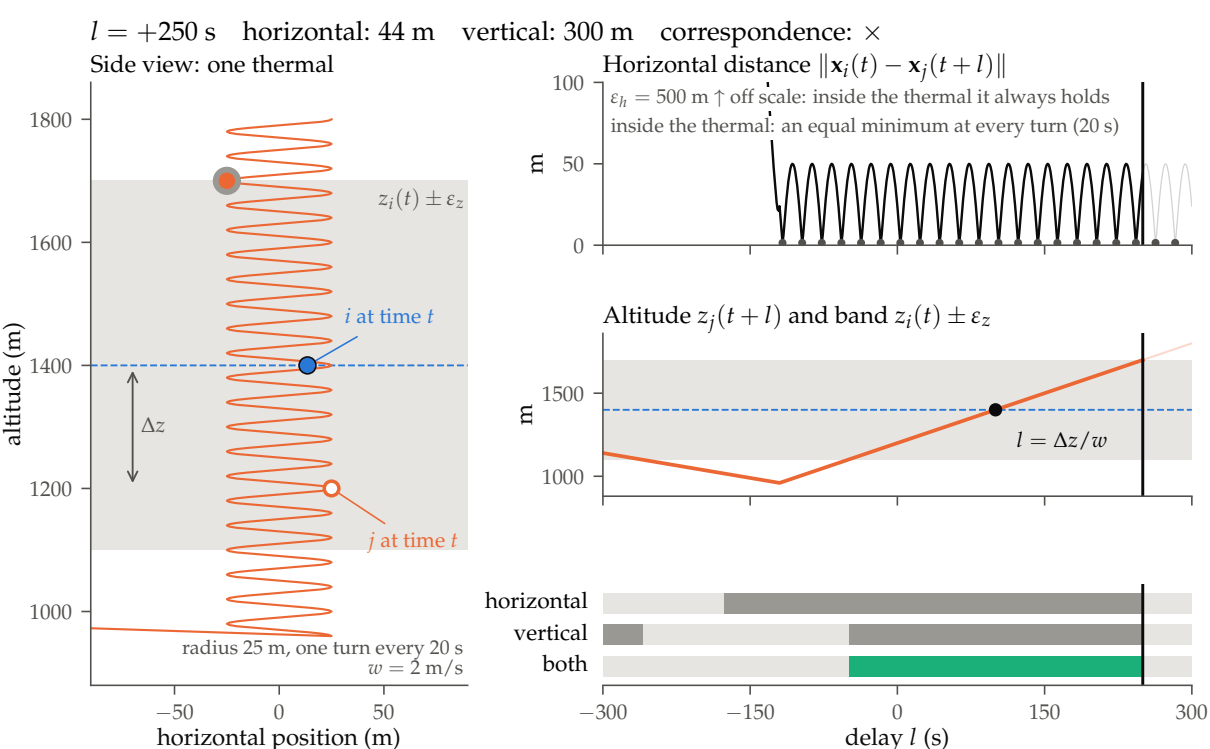

$l = +200$ s horizontal: 24 m vertical: 200 m correspondence: ✓
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

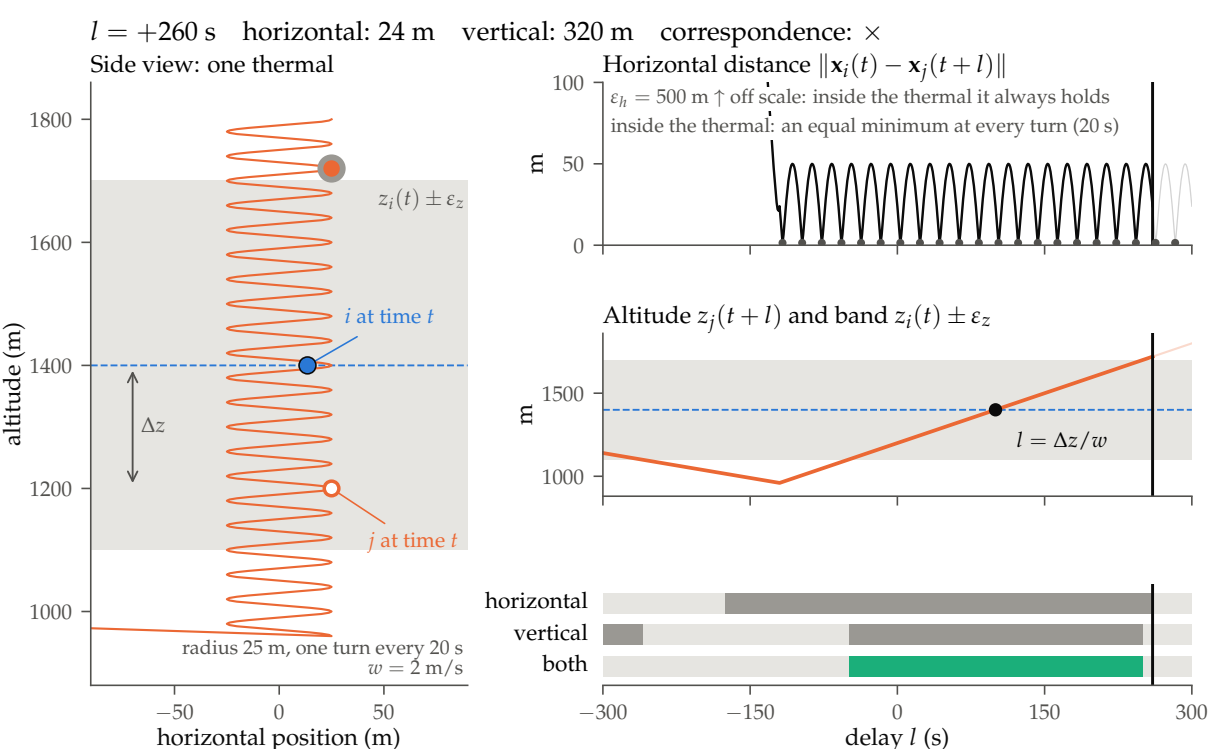

$l = +210$ s horizontal: 44 m vertical: 220 m correspondence: ✓
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

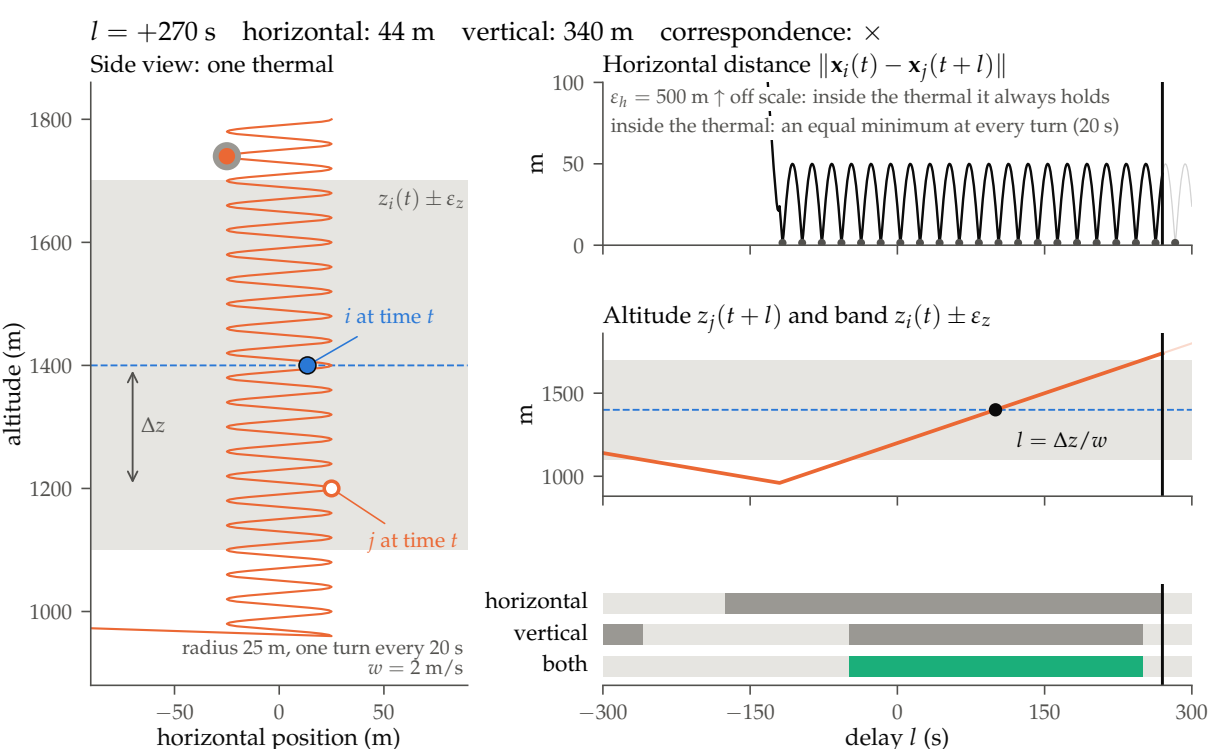

$l = +220$ s horizontal: 24 m vertical: 240 m correspondence: ✓
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

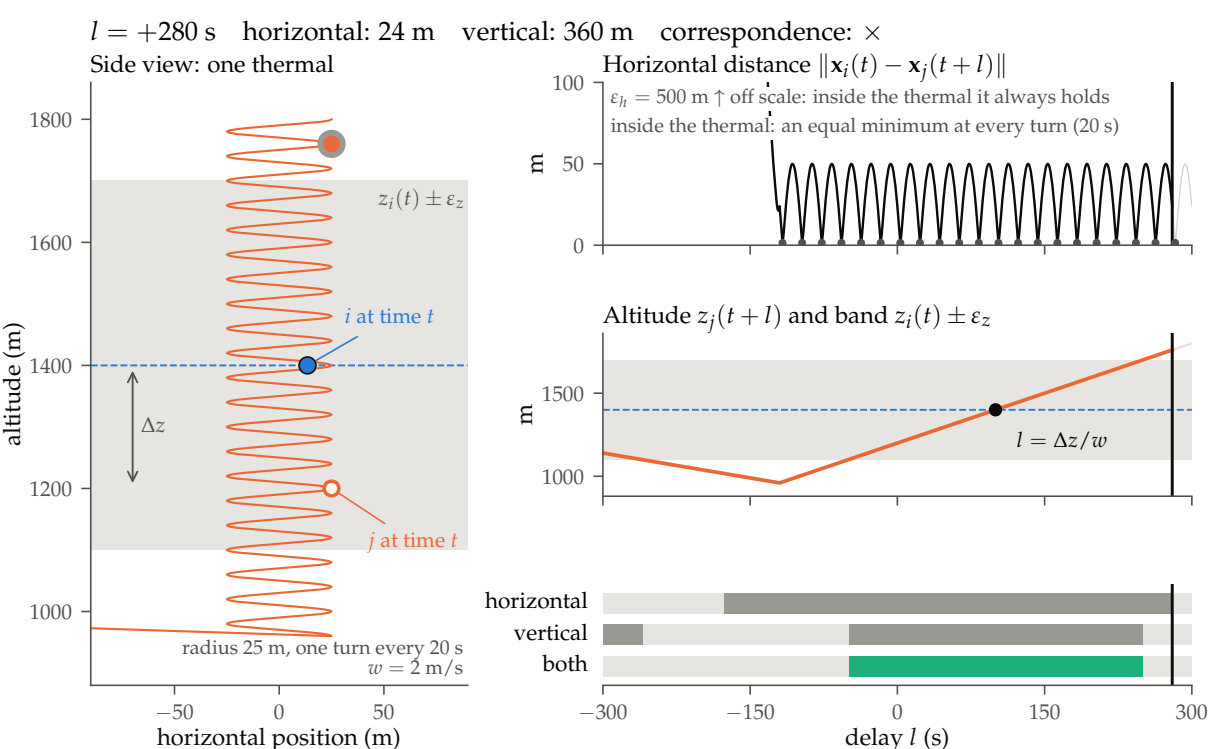

$l = +230$ s horizontal: 44 m vertical: 260 m correspondence: ✓
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

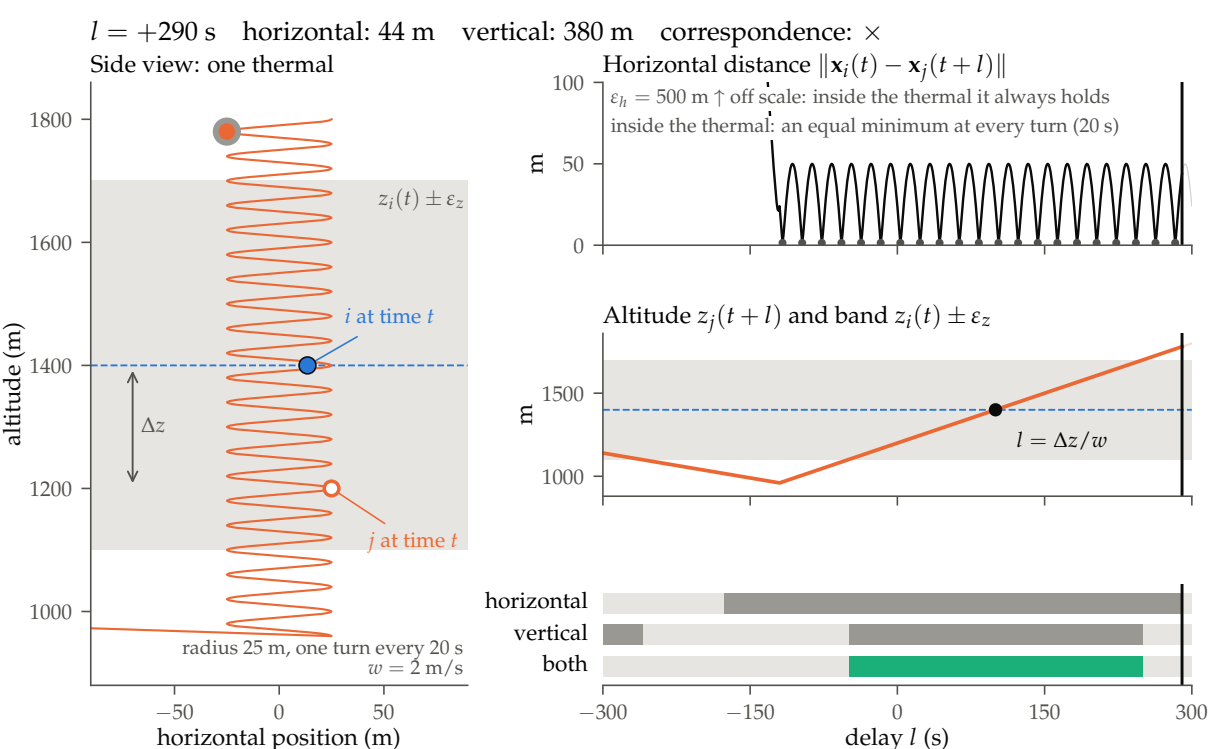

$l = +240$ s horizontal: 24 m vertical: 280 m correspondence: ✓
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

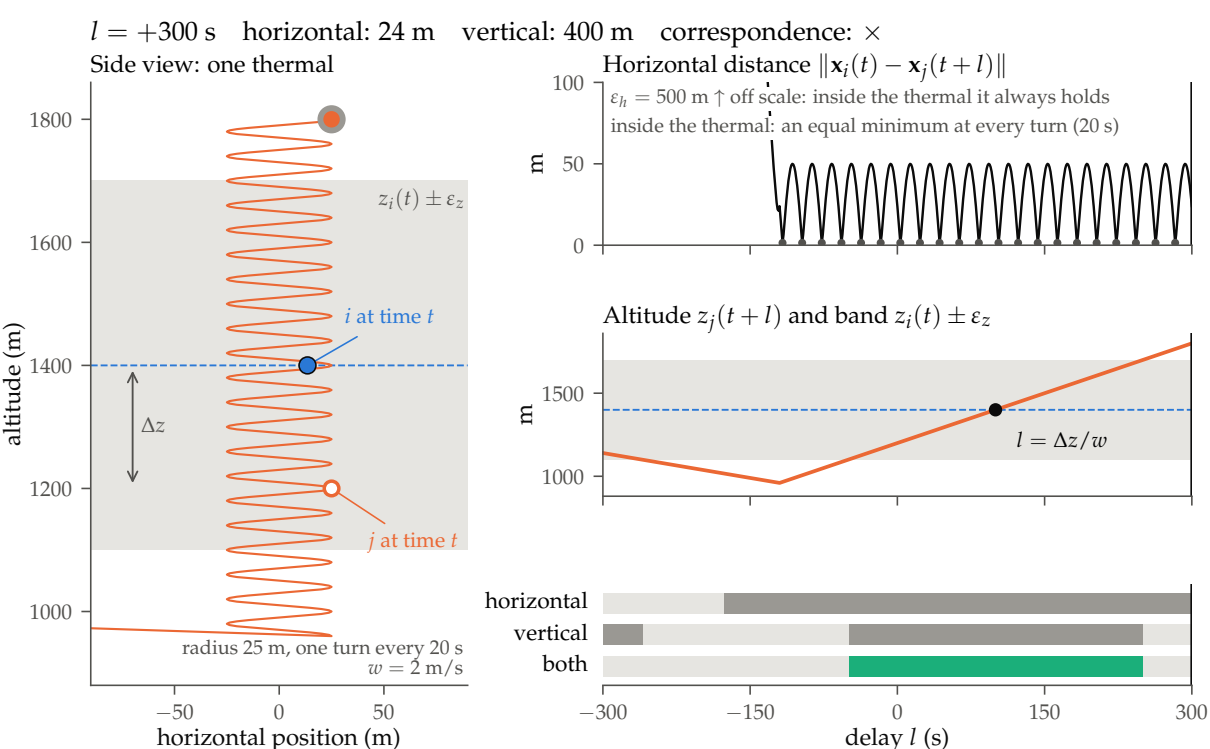

$l = +250$ s horizontal: 44 m vertical: 300 m correspondence: ×
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)

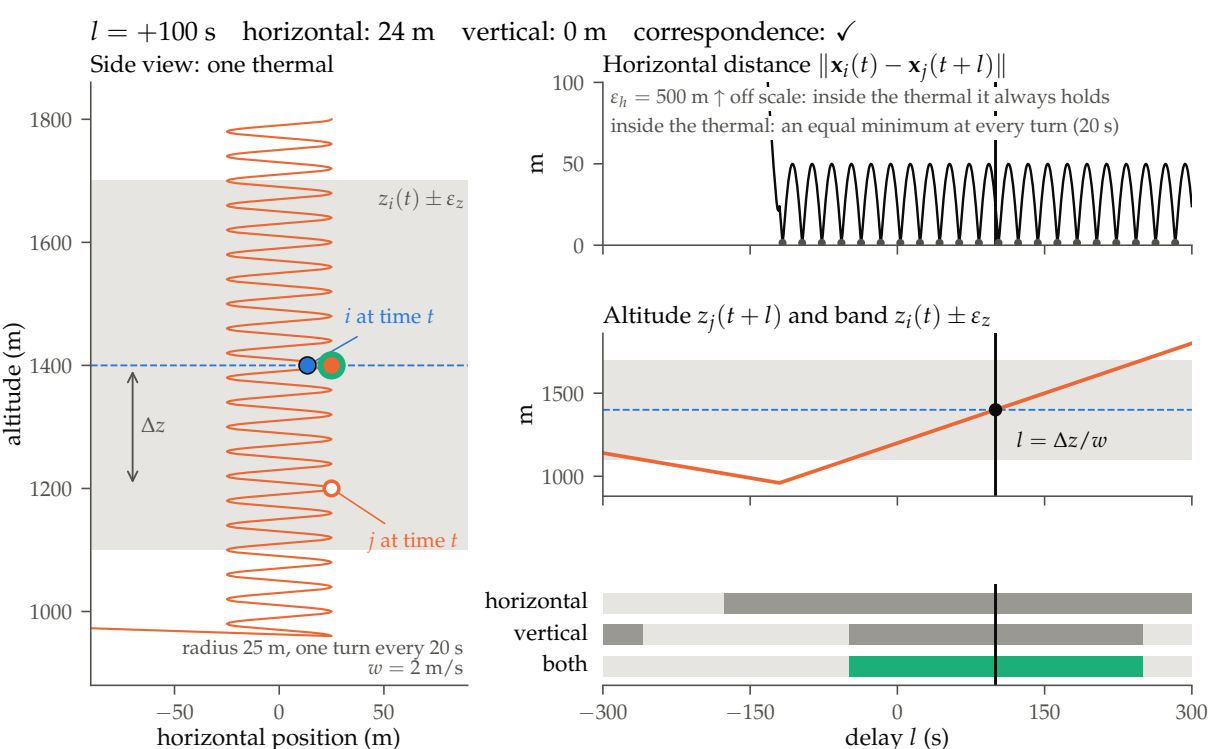

$l = +260$ s horizontal: 24 m vertical: 320 m correspondence: ×
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)


$l = +270$ s horizontal: 44 m vertical: 340 m correspondence: ×
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)


$l = +280$ s horizontal: 24 m vertical: 360 m correspondence: ×
Side view: one thermal
altitude (m)
1800
1600
1400
1200
1000
$z_i(t) \pm \varepsilon_z$
i at time t
j at time t
Δz
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)


$l = +290$ s horizontal: 44 m vertical: 380 m correspondence: ×
Side view: one thermal
altitude (m)
1800
1600
1400
1200
1000
$z_i(t) \pm \varepsilon_z$
i at time t
j at time t
Δz
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)


$l = +300$ s horizontal: 24 m vertical: 400 m correspondence: ×
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)


$l = +100$ s horizontal: 24 m vertical: 0 m correspondence: ✓
Side view: one thermal
1800
1600
1400
1200
1000
altitude (m)
$z_i(t) \pm \varepsilon_z$
i at time t
Δz
j at time t
radius 25 m, one turn every 20 s
$w = 2$ m/s
−50
0
50
horizontal position (m)
Horizontal distance $\|\mathbf{x}_i(t) - \mathbf{x}_j(t+l)\|$
100
50
0
m
$\varepsilon_h = 500$ m ↑ off scale: inside the thermal it always holds
inside the thermal: an equal minimum at every turn (20 s)
Altitude $z_j(t+l)$ and band $z_i(t) \pm \varepsilon_z$
1500
1000
m
$l = \Delta z / w$
horizontal
vertical
both
−300
−150
0
150
300
delay l (s)